

Canonical Forms Need Not Be Subgraphs

**Rigidity and sparsity of surreal option graphs:
two five-vertex counterexamples, sharp graph minima, minimal-form
enumeration, an exact girth–size tradeoff, and planar high-girth forms**

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Abstract

A question explicitly posed by Matthew Roughan in 2018 asks whether the canonical representation of a surreal number is a subgraph of every other representation of that number. We give a negative answer using a five-vertex form of $1/2$: its underlying undirected graph is a pentagon, while the canonical graph is a triangle. The obstruction therefore persists after all edge directions, left/right designations, roots, and numerical labels are forgotten. Five vertices and form rank three are both smallest possible for such a counterexample. A second, structurally different five-vertex witness, whose intermediate values are only 0 and 1, is recorded beside it; an exhaustive search shows that there are exactly sixteen five-vertex counterexamples, of which eight are triangle-free and four have value $1/2$.

The negative answer leads to positive structural results in two opposite directions. On the rigid side, every finite representation contains a directed path through all sign-prefix values of its value; for a dyadic x with birthday b and reduced denominator 2^k , the minimum vertex count, edge count, and cycle rank are respectively $b + 1$, $b + k$, and k ; there are exactly $2^{b(b-1)/2-k}$ vertex-minimal forms, all containing the canonical graph as a value- and side-preserving spanning subgraph; and the canonical form is the unique edge-minimizer. On the sparse side, every finite-birthday surreal value has representations of arbitrarily large girth, obtained here by two independent constructions: a spaced-spine construction with exact size, rank, and cycle-rank formulas, and a tree-and-spine construction with integer-valued markers. Both can be made planar with maximum degree at most three. For the single value $1/2$ the exact girth–size tradeoff is determined: the minimum number of vertices at girth at least g is 3, 5, and g for $g = 3$, $g = 4$, and $g \geq 5$, and every cycle length except four is realized exactly. Consequently the values whose canonical graphs occur in every representation are exactly the integers, both in the weakest unlabelled sense and in a stronger rooted, directed, L/R -preserving sense. Two disjoint transfinite results are proved at ω : a limit-stage obstruction to choosing all prefix representatives coherently, and a triangle-free representation of ω with unchanged birthday. Exact-arithmetic code exhausts all 3,860 forms with at most five vertices and checks them with three oracles of different kinds.

Status and scope. The elementary counterexamples and the general theorems below are accompanied by complete mathematical proofs. This is an unrefereed, AI-assisted research draft, not a claim of machine-checked formalization or independently certified priority. A targeted literature search on the date above found the original question and later related work, but no subsequent resolution. Search coverage and publisher access were limited, and this absence is not a certificate of novelty. Bibliographic metadata for the 2023 journal version of the source was located, but its full text was not retrievable; no claim is made that the 2018 wording persists unchanged there. The result concerns dependency graphs of *forms*, not graphs obtained by identifying every pair of equal-valued vertices.

Provenance. This article merges two independently prepared research packages on the same question, `canonical-forms-need-not-be-subgraphs` and `sparse-option-graphs`. Both answered the question negatively, at the same weakest reading, with a five-vertex form of $1/2$ whose underlying graph is a pentagon; both used the same conventions (structurally distinct raw forms as vertices, the reduced Dali canonical map, the dependency DAG with unique sink O). The two witnesses, however, are genuinely different forms, not renamings of one another: the first hangs its long arm on the left of the root and passes through a vertex of value -1 , and has form rank three; the second hangs its long arm on the right, uses only the values 0 and 1 , has the empty form as a direct option of the root, and has form rank four. Both are printed in full in [section 3](#). Beyond the shared pentagon the two packages diverge almost completely: the first supplies the prefix theory, exact graph minima, the classification and count of vertex-minimal forms and the minimality of the counterexample; the second supplies the planar degree-three high-girth construction, the full cycle spectrum and exact extremal function at $1/2$, the stronger rooted-colour classification, and the high-girth form of ω . Where the two packages proved the same statement by different routes, both proofs are retained and labelled.

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1 The question and the answer

The distinction between a number and a particular expression for that number is especially important for surreal numbers. A form is a recursively specified pair of sets of earlier forms. Several forms can have the same numerical value while retaining different dependency graphs. This makes it natural to ask whether a canonical representation is an unavoidable graph-theoretic core.

In the caption to Figure 3 of *Surreal Birthdays and Their Arithmetic*, Roughan asks the following question explicitly [2, p. 7]:

“is the canonical form always a subgraph of any alternative form of the same number?”

The paper distinguishes identity of forms from equality of values and uses the reduced dyadic, or Dali, canonical representation. We use precisely that convention. The problem is a small, concrete question about representation structure, rather than a major foundational conjecture about the surreal field. Its advantage as a research target is that a finite counterexample can settle the universal assertion without any numerical approximation.

The question is ambiguous about what an embedding must preserve. There are at least three reasonable readings: actual occurrence of the canonical form as a subform; containment with numerical and left/right labels preserved; and containment of the underlying unlabelled undirected graph. The first implies the second and the second the third, so refuting the weakest reading refutes all three at once. That is what the counterexamples below do. They say nothing about containment as a graph minor or after replacing edges by paths.

Here is the entire counterexample. The symbols on the left name *forms*:

$$O = \{ \mid \}, \quad P = \{ O \mid \}, \quad N = \{ \mid O \}, \quad Z = \{ N \mid \}, \quad F = \{ Z \mid P \}. \quad (1.1)$$

Their values, in order, are 0, 1, −1, 0, 1/2. The graph of F consists of the two paths

$$F \longrightarrow Z \longrightarrow N \longrightarrow O, \quad F \longrightarrow P \longrightarrow O. \quad (1.2)$$

It is a pentagon after orientations are forgotten. In contrast, $C(1/2) = \{ O \mid P \}$ has all three edges of a triangle. A pentagon has no triangle as a subgraph.

A second, structurally different pentagon witnesses the same failure:

$$O = \{ \mid \}, \quad P = \{ O \mid \}, \quad Z' = \{ \mid P \}, \quad P' = \{ Z' \mid \}, \quad G = \{ O \mid P' \}, \quad (1.3)$$

with values 0, 1, 0, 1, 1/2. Neither witness is a renaming, a negation, or an automorphic image of the other: (1.1) contains a vertex of value −1 and its root has no edge to O , while (1.3) uses only the values 0 and 1 and its root has O as a direct option. We keep (1.1) as the headline because it is doubly extremal — five vertices and form rank three are both least possible — and record (1.3) beside it because it buys a witness free of negative intermediate values, and because it is the first member of a second infinite cycle family (section 7). An exhaustive search shows there are exactly sixteen five-vertex counterexamples; these two are different members of that certified list.

The later results explain why this failure is not a coincidence and what survives in place of the conjectured containment. For a finite form A let $v(A)$ and $e(A)$ be the numbers of distinct subforms and option edges, and let

$$\beta(A) = e(A) - v(A) + 1 \quad (1.4)$$

be the cycle rank of its connected underlying graph. If $x \in \mathbb{D}$ has reduced denominator 2^k and intrinsic birthday b , the results are

$$\min_{\text{val}(A)=x} v(A) = b + 1, \quad \min_{\text{val}(A)=x} e(A) = b + k, \quad \min_{\text{val}(A)=x} \beta(A) = k, \quad (1.5)$$

$$\#\{A : \text{val}(A) = x, v(A) = b + 1\} = 2^{b(b-1)/2-k}. \quad (1.6)$$

The minima are over finite numerical forms. These are exact statements about structural forms, not about syntactic descriptions with arbitrary vertex names.

The principal global conclusion is a complete classification in the weakest usual sense of subgraph containment:

$$\left[U(C(x)) \text{ embeds as a subgraph of } U(A) \text{ for every finite } A \text{ with } \text{val}(A) = x \right] \iff x \in \mathbb{Z}. \quad (1.7)$$

Here U forgets directions, option sides, roots, and value labels. We do not silently replace “subgraph” by “minor,” “subdivision,” or “reachability.” Those are different questions. A second classification, proved by an independent argument in [section 10](#), gives the same answer under a strictly stronger containment convention that preserves the root, the arrow directions, and the L/R edge labels.

The flexibility behind (1.7) is quantitative. Every dyadic value has finite representations of arbitrarily large girth, and they can be taken planar with maximum degree at most three:

$$\text{for every } x \in \mathbb{D} \text{ and } g \geq 3 \text{ there is a finite } X \text{ with } \text{val}(X) = x, U(X) \text{ planar, } \Delta \leq 3, \text{ girth} \geq g. \quad (1.8)$$

For the single value $1/2$ the tradeoff is known exactly. Writing $m_{1/2}(g)$ for the fewest distinct subforms in a form of value $1/2$ whose graph has girth at least g ,

$$m_{1/2}(g) = \begin{cases} 3, & g = 3, \\ 5, & g = 4, \\ g, & g \geq 5, \end{cases} \quad (1.9)$$

and every cycle length $s = 3$ or $s \geq 5$ occurs as the entire underlying graph of some form of value $1/2$, while C_4 occurs at no noninteger value at all.

These results explain why canonical subgraphs can disappear: the recursive identities let numerical information be preserved while the shared dependencies of a form are moved arbitrarily far apart.

2 Forms, values, and dependency graphs

2.1 Numerical forms and structural identity

Definition 2.1. A *finite numerical form* is a hereditarily finite ordered pair $A = \{L_A \mid R_A\}$ of sets of numerical forms, subject to

$$\text{val}(B) < \text{val}(D) \quad (B \in L_A, D \in R_A). \quad (2.1)$$

Its value is the simplest surreal number strictly above every left-option value and strictly below every right-option value. The initial form is $O = \{ \mid \}$ and has value zero.

The standard simplicity theorem ensures that this definition agrees with the usual numerical interpretation of surreal cuts. A convenient account using sign expansions is given in [1, Sections 2–3]; a self-contained treatment of the simplicity theorem itself is given by Schleicher and Stoll [5, Section 5.1]. Every descendant of a numerical form is required to be a numerical form; general combinatorial games are not admitted. For a finite form the value is a dyadic rational. Throughout this article, a *finite value* means a value of finite birthday, not an arbitrary surreal bounded in absolute value by an integer. For example,

$$\text{val}(\{C(-1) \mid \}) = 0, \quad \text{val}(\{C(0) \mid C(2)\}) = 1. \quad (2.2)$$

These cuts are not interpreted by taking averages in general: the rule is simplicity, not the arithmetic midpoint rule. Midpoints do arise for the specific adjacent dyadic bounds in the canonical construction.

Two forms are identical exactly when their left sets are identical and their right sets are identical, recursively. In particular,

$$\{C(-1) \mid\} \neq O \quad \text{as forms, although both have value 0.} \quad (2.3)$$

No additional labels or tags are allowed to make two structurally identical forms different. This last restriction matters in the large-girth construction: we must prove that apparently different vertices really are different forms.

Definition 2.2. The *form rank* is

$$\rho(O) = 0, \quad \rho(A) = 1 + \max\{\rho(B) : B \in L_A \cup R_A\} \quad (A \neq O). \quad (2.4)$$

The *intrinsic birthday* $b(x)$ belongs to the numerical value x : it is the length of its sign expansion, equivalently the least possible form rank of a representation of x .

Thus $b(0) = 0$ but the form Z in (1.1) has rank two, and $\rho(G) = 4$ in (1.3) although $b(1/2) = 2$. Confusing these two quantities would obscure both the counterexample and the extremal results: a graph statement about raw forms cannot be inferred from the birthday of the value alone.

2.2 The graph convention

Definition 2.3. The *dependency graph* $D(A)$ has one vertex for each distinct form in the transitive option closure of A , including A itself. It has a directed edge $B \rightarrow D$ of type L when $D \in L_B$ and of type R when $D \in R_B$. The root is A . The underlying simple undirected graph is denoted by $U(A)$.

Every option edge strictly decreases form rank, so the directed graph is acyclic. The root reaches every vertex. The only vertex with no outgoing edge is O , since an empty left set and an empty right set uniquely specify O . Consequently $U(A)$ is connected. A child cannot occur on both sides of a numerical cut, because that would require its value to be strictly less than itself. There are no loops or parallel undirected edges.

An undirected graph has *girth* equal to the length of a shortest simple cycle, and infinite girth when it has no cycle. Throughout, subgraph containment is ordinary and non-induced: removing edges is permitted, contracting them is not.

We distinguish three increasingly weaker demands: literal containment of canonical subforms; a rooted, value- and L/R -preserving graph embedding; and an arbitrary unlabelled undirected subgraph embedding. The counterexamples fail even the third demand. Some later positive statements use the second demand, with vertices identified by their values, not by literal identity of forms.

Remark 2.4 (Why not identify equal values?). Merging Z with O in (1.1) changes the graph, not merely its notation. Edges and cycles are altered. The question under consideration uses forms as vertices, as does Roughan's discussion of two distinct zero-valued vertices [2, Section 2]. An argument about a quotient graph of values would address another problem, and a sharper objection is available: a numerical quotient need not even be acyclic. In (1.3) the edges $P \rightarrow O$ and $Z' \rightarrow P$ would become $1 \rightarrow 0$ and $0 \rightarrow 1$, so the quotient of the pentagon is not a directed acyclic graph at all, hence not an option graph in the sense of this article.

The opposite error must also be avoided. Making several separately allocated copies of the empty form does not create several mathematical vertices: all such copies are the same ordered pair of empty sets, hence the same form. Neither witness relies on that implementation error. Their repeated values occur in genuinely different forms, separated by different form ranks.

2.3 The reduced dyadic canonical form

Write $\mathbb{D} = \mathbb{Z}[1/2]$. For $x \in \mathbb{D}$, let $\kappa(x) = k$ when the positive reduced denominator is 2^k ; in particular $\kappa(n) = 0$ for integers. Define

$$C(x) = \begin{cases} O, & x = 0, \\ \{C(n-1) \mid\}, & x = n \in \mathbb{Z}_{>0}, \\ \{\mid C(n+1)\}, & x = n \in \mathbb{Z}_{<0}, \\ \{C((m-1)/2^k) \mid C((m+1)/2^k)\}, & x = m/2^k, \quad k > 0, \quad m \text{ odd}. \end{cases} \quad (2.5)$$

This is the Dali construction [6, Eq. (2.71)]. All fractions on the right are reduced before the recursion. In particular $C(n)$ is a chain of $|n|$ unary steps from zero, whereas $C(1/2) = \{O \mid C(1)\}$ is a binary cut.

That (2.5) does define a form of value x is an induction. The integer clauses have the stated values by simplicity. In the fourth clause the two options are the neighbouring points of the 2^{-k} grid, and $m/2^k$ is the unique dyadic of least denominator exponent strictly between them: no number of denominator exponent below k lies in an open interval of length 2^{1-k} centred at a point of exponent exactly k , and among the numbers of exponent k only $m/2^k$ itself lies there. Hence $\text{val}(C(x)) = x$ throughout.

There is a terminological subtlety. Another standard canonical cut includes *all* shorter sign prefixes on their appropriate sides. That cut can have more edges than (2.5). We reserve $C(x)$ for the reduced Dali form. The initial half-integer counterexample does not depend on this distinction, since both conventions give the same three-vertex graph at $1/2$. The edge minima, uniqueness, and enumeration results below specifically concern (2.5).

3 Two five-vertex obstructions

3.1 The rank-minimal witness

Theorem 3.1 (A pentagon represents $1/2$). *The form F in (1.1) has value $1/2$, rank three, five vertices, and five edges. Its underlying graph is C_5 . The underlying graph of $C(1/2)$ is K_3 , so it is not a subgraph of $U(F)$.*

Proof. The empty cut has value zero. The simplest values strictly above and strictly below zero are respectively 1 and -1 , giving the stated values of P and N . Since zero is already strictly greater than -1 , the cut $Z = \{N \mid\}$ has value zero. The simplest value strictly between zero and one is $1/2$, so $\text{val}(F) = 1/2$.

The ranks of O, P, N, Z, F are respectively 0, 1, 1, 2, 3. The different ranks separate all potentially equal-valued vertices, and P and N are distinct because their values differ. The complete edge set is exactly (1.2). In the undirected graph the cyclic order is F, Z, N, O, P, F , with no chord. Meanwhile $C(1/2) = \{O \mid P\}$ has edges to both O and P , and P has an edge to O . A triangle cannot embed into a chordless pentagon. $\square$

The distinction is combinatorial rather than an artifact of edge directions or numerical labels. No selection of three vertices in the alternative graph has all three required edges. Also, the assertion is about ordinary, non-induced subgraphs: permitting extra edges in a host would not help when the required triangle is absent.

This counterexample does *not* refute a minor version of the question. A pentagon has a triangle as a minor and is a subdivision of a triangle. Keeping these relations separate is essential when looking for a corrected statement.

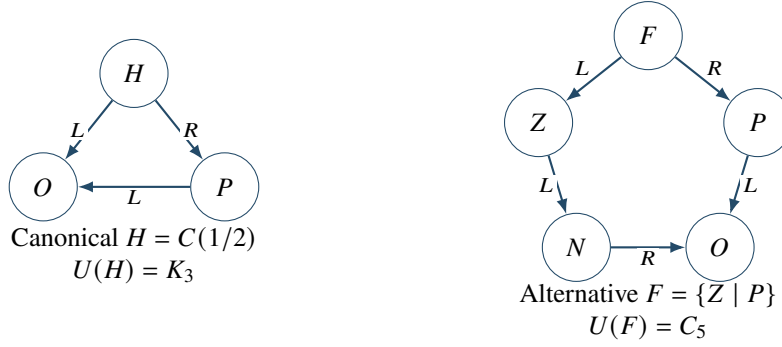


Figure 1: The canonical triangle and the five-vertex obstruction. Vertex labels name forms. In the right-hand graph O and Z both have value zero but are not identical forms. Directions point from a form to an option.

3.2 A second witness with no negative value

The next theorem records an independently discovered witness. It is not a relabelling of [theorem 3.1](#): what it buys is a counterexample whose intermediate values are only 0 and 1, so that no negative number is imported at all, and whose root has the empty form itself as a direct option.

Theorem 3.2 (A second pentagon of value $1/2$). *The form G in (1.3) has value $1/2$, rank four, five vertices, and five edges, and $U(G) = C_5$, drawn in [figure 2](#). Hence $U(C(1/2)) = K_3$ is not a subgraph of $U(G)$. The forms F of (1.1) and G of (1.3) are distinct, and neither is the negation of the other.*

Proof. The empty cut has value zero, so $\text{val}(P) = 1$: one is the simplest number strictly above zero. Zero is the simplest number strictly below one, so $\text{val}(Z') = 0$, and then $\text{val}(P') = 1$ by the same computation as for P . Finally the simplest number strictly between $\text{val}(O) = 0$ and $\text{val}(P') = 1$ is $1/2$, so $\text{val}(G) = 1/2$. Each cut is valid: the unary cuts impose no left-versus-right comparison, and the final cut satisfies $0 < 1$.

The ranks of O, P, Z', P', G are 0, 1, 2, 3, 4, so all five forms are structurally distinct even though two pairs share a value. The complete edge list is

$$P \xrightarrow{L} O, \quad Z' \xrightarrow{R} P, \quad P' \xrightarrow{L} Z', \quad G \xrightarrow{L} O, \quad G \xrightarrow{R} P', \quad (3.1)$$

and there are no others. Forgetting directions gives exactly the chordless pentagon $G - O - P - Z' - P' - G$, which has no triangle.

For the last assertion, F has rank three and G rank four, so they are distinct. The negation of G has value $-1/2$, not $1/2$. More structurally, $D(F)$ has a vertex of value -1 and $D(G)$ does not, and the root of $D(G)$ has an edge to O while the root of $D(F)$ does not. $\square$

3.3 Exactly sixteen five-vertex counterexamples

The two witnesses are not isolated, and they are not the only ones. The exhaustive computation described in [section 12](#) finds that among all 3,860 structurally distinct forms with at most five vertices there are exactly sixteen whose underlying graph fails to contain its canonical underlying graph, all of them at five vertices. Eight of the sixteen are triangle-free, and four of those have value $1/2$. The form F of (1.1) is the entry of rank three and girth five; the form G of (1.3) is a different entry, of rank four and girth five. We do not claim either is unique. The reconciliation of the two failure counts is discussed in [section 12](#): the sixteen include girth-three forms of $\pm 1/4$ and $\pm 3/4$, whose canonical graph is a diamond needing two triangles, and restricting the sixteen to noninteger values of girth at least four returns exactly the eight triangle-free forms.

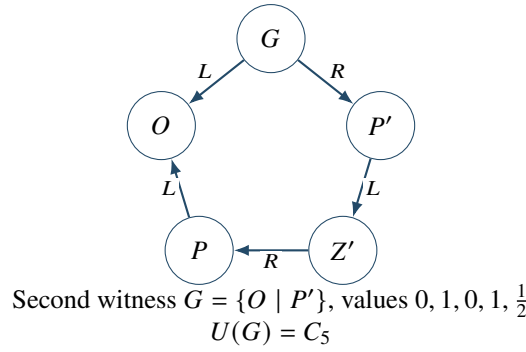


Figure 2: The second five-vertex counterexample. Here O and Z' are distinct zero-valued forms and P and P' are distinct one-valued forms; no negative value occurs. Compare [figure 1](#), whose pentagon passes through a vertex of value -1 and whose root is not adjacent to O .

3.4 An independent order-theoretic verification

Neither conclusion depends on a heuristic reading of braces. For number forms the recursive comparison relation can be written without any reference to rational values:

$$X \leq Y \iff (\forall X^L : \neg(Y \leq X^L)) \wedge (\forall Y^R : \neg(Y^R \leq X)). \quad (3.2)$$

The accompanying checker implements (3.2) using only option identifiers, never cached rationals. The bundled `verify_sparse.py` suite applies it to G and $C(1/2)$: it verifies both $G \leq C(1/2)$ and $C(1/2) \leq G$, and checks all 36 ordered comparisons among their six distinct subforms against exact rational ordering. This is a structurally different check from the second cut evaluator of [section 12](#), which re-derives values but still works with numbers; (3.2) never computes a value at all. We use (3.2) again as a proof tool in [section 10](#).

4 Canonical prefixes cannot disappear as values

The edges of a canonical representation are not unavoidable, but its simpler prefix values are. We begin with the order-theoretic fact that makes this precise.

4.1 Sign expansions and convexity

A surreal sign expansion is a sequence of $+$ and $-$ indexed by an ordinal. The order is lexicographic, with an absent sign lying between $-$ and $+$. We write $q \preccurlyeq y$ when the sign expansion of q is an initial segment of that of y , allowing equality. Proper initial segments are simpler. These standard facts and their relationship with cuts are developed in [[1](#), Sections 2–3].

Lemma 4.1 (Convexity of a simplicity cone). *For any surreal q , the class*

$$I(q) = \{y : q \preccurlyeq y\} \quad (4.1)$$

is order-convex. In particular, if $q \preccurlyeq y$ and z lies between q and y , then $q \preccurlyeq z$.

Proof. Suppose a, c extend the sign expansion of q and $a < z < c$. If z fails to extend q , let α be the first position before the end of q at which z either terminates or differs from q . At that position a and c have the same sign as q . Lexicographic comparison then places z on the same strict side of both a and c , contradicting $a < z < c$. $\square$

Lemma 4.2 (Descent toward a prefix). *Let A be a numerical form, $y = \text{val}(A)$, and $q \prec_s y$. If $q < y$, there is a left option B with*

$$q \leq \text{val}(B) < y, \quad q \preceq_s \text{val}(B). \quad (4.2)$$

If $y < q$, there is a right option B with

$$y < \text{val}(B) \leq q, \quad q \preceq_s \text{val}(B). \quad (4.3)$$

Proof. The value q cannot satisfy all the strict inequalities defining the cut of A : if it did, y would not be the simplest element of that interval, because q is a proper sign prefix of y . Suppose $q < y$. Every right-option value is strictly greater than y , so no right option can exclude q . Some left option must therefore have value at least q . Every left-option value is strictly less than y , giving the required interval. Apply Lemma 4.1. The case $y < q$ is the left/right reversal. $\square$

Theorem 4.3 (Prefix occurrence). *If A is a well-founded numerical form and $q \preceq_s \text{val}(A)$, then a descendant subform of A has value q . It can be reached by a finite directed path. When $q < \text{val}(A)$, the path can use only left-option edges; when $q > \text{val}(A)$, it can use only right-option edges.*

Proof. Use induction on the well-founded rank of A . If $q = \text{val}(A)$, the length-zero path suffices. Otherwise Lemma 4.2 supplies an option B of smaller rank with $q \preceq_s \text{val}(B)$. Apply the induction hypothesis to B and prepend the edge $A \rightarrow B$. In the case $q < \text{val}(A)$ all subsequent values remain at least q , so every nonterminal step uses a left option. The opposite case uses right options throughout. $\square$

The proof is deliberately a rank induction, not a compactness argument. It works for arbitrary set-sized well-founded forms as well as finite ones. We return to the transfinite case in section 11.

4.2 Finite prefixes and the canonical dependency graph

For a finite-birthday value x , list its sign-prefix values by length:

$$p_0 = 0, p_1, \dots, p_b = x, \quad b = b(x). \quad (4.4)$$

There is exactly one prefix at each length, although the values need not be in increasing numerical order. For example, the prefixes of $5/8$ are

$$0, 1, \frac{1}{2}, \frac{3}{4}, \frac{5}{8}. \quad (4.5)$$

Lemma 4.4 (The finite canonical prefix chain). *Let $x \in \mathbb{D}$ and $k = \kappa(x)$. Then*

$$b(x) = \lceil |x| \rceil + k. \quad (4.6)$$

The vertices of $D(C(x))$ are exactly $C(p_0), \dots, C(p_b)$. For each $j > 0$, its options are the nearest earlier prefix below p_j and the nearest earlier prefix above p_j , omitting a side when no such prefix exists. One of these options is $C(p_{j-1})$. Exactly k of $p_1, \dots, p_b$ are nonintegers.

Proof. For a positive integer n , the prefixes are $0, 1, \dots, n$, and the assertions follow directly from (2.5). Suppose x is positive and nonintegral, with $n < x < n + 1$ and $n \geq 0$. Its initial prefixes are $0, 1, \dots, n + 1$. Thereafter its prefixes are obtained by binary search in the interval $(n, n + 1)$: first test the midpoint, retain the half containing x , and repeat until the midpoint is x .

At each step the current endpoints are precisely the nearest earlier prefix values below and above the next midpoint. If that midpoint first occurs with reduced denominator 2^r , the endpoints are its two adjacent multiples on the same 2^{-r} grid. Hence they are exactly the two options in (2.5), after reducing their fractions. One endpoint is the immediately previous midpoint, except at the first midpoint, when the previous prefix is $n + 1$. Thus consecutive prefixes are linked by canonical option edges.

A reduced denominator 2^k is first reached on the k th midpoint step. There are $n + 1$ initial nonzero integer prefixes followed by k noninteger prefixes. This gives $b = n + 1 + k$, the vertex description, and all option claims. Negation reverses the signs and exchanges option sides, proving the statements for negative values. Zero is immediate. $\square$

Remark 4.5 (A second route to the birthday formula). The identity (4.6) alone, without the prefix-chain structure, also follows by induction on k directly from (2.5), and the second route is worth recording because it is what one needs when only the height of the canonical form matters. Integer values follow directly from their unary chains; negation preserves rank, so it remains to consider $x \in (n, n + 1)$ with $n \geq 0$ and $\kappa(x) = k \geq 1$. For $k = 1$ the two canonical options are $C(n)$ and $C(n + 1)$, of ranks n and $n + 1$, giving $\rho(C(x)) = n + 2$. For $k > 1$, exactly one of the two neighbouring grid points $(m \pm 1)/2^k$ has reduced denominator exponent $k - 1$, since exactly one of $m - 1, m + 1$ has 2-adic valuation one; the other has exponent at most $k - 2$ or is an integer. Both options stay in $[n, n + 1]$, so by induction their ranks are at most $n + k$ with the maximum $n + k$ attained, and $\rho(C(x)) = n + k + 1$. Since $b(x) = \rho(C(x))$ and $\lceil x \rceil = n + 1$, this is (4.6). The same 2-adic observation reappears in Proposition 8.5.

Theorem 4.6 (A path through every finite prefix). *If A is a finite form of value x with prefixes (4.4), then $D(A)$ contains a directed path on which the values*

$$p_b, p_{b-1}, \dots, p_0 \quad (4.7)$$

occur in that order. Additional vertices may occur between consecutive listed values. In particular,

$$v(A) \geq b(x) + 1, \quad \rho(A) \geq b(x). \quad (4.8)$$

Proof. Starting at A , apply theorem 4.3 to p_{b-1} . From the resulting form of value p_{b-1} , apply it to p_{b-2} , and continue. This is a finite concatenation. Ranks strictly decrease along every edge, so the concatenation cannot repeat a vertex. Each of the $b + 1$ distinct prefix values therefore appears on a single directed path, whose length is at least b . $\square$

This is the basic positive replacement for literal subgraph containment. If only reachability is required, the canonical prefix vertices can be represented coherently in the finite case. The paths that realize different canonical edges may overlap. The theorem consequently does not assert a topological-minor embedding, for which appropriate internal vertex-disjointness would be needed.

5 Sharp minima and an exact enumeration

5.1 The denominator as minimum cycle rank

Lemma 5.1 (One-sided finite cuts have integer values). *If a finite numerical form has a noninteger value, both its left and right option sets are nonempty. In particular, it has at least two outgoing edges.*

Proof. If both sets are empty, its value is zero. If only the left set is nonempty, let a be its largest option value. When $a < 0$, zero is allowed and is simplest. When $a \geq 0$, the smallest integer strictly greater than a is the simplest surreal strictly greater than a . Thus the value is an integer. An empty left set is handled by negation. $\square$

Theorem 5.2 (Exact graph minima). *Let $x \in \mathbb{D}$, $b = b(x)$, and $k = \kappa(x)$. Every finite form A of value x satisfies*

$$v(A) \geq b + 1, \quad \beta(A) \geq k, \quad e(A) \geq b + k. \quad (5.1)$$

The canonical form attains all three equalities simultaneously.

Proof. The vertex bound is [theorem 4.6](#). Since O is the unique sink,

$$\begin{aligned}\beta(A) &= \sum_{B \in V(D(A))} \deg^+(B) - v(A) + 1 \\ &= \sum_{B \in V(D(A)) \setminus \{O\}} (\deg^+(B) - 1).\end{aligned}\tag{5.2}$$

Every summand is nonnegative. By [Lemma 4.4](#), exactly k of the canonical prefix values are nonintegral. The prefix-occurrence theorem provides distinct vertices with those values. Each has outdegree at least two by [Lemma 5.1](#), so these k vertices contribute at least k to (5.2). This proves $\beta(A) \geq k$. Finally,

$$e(A) = v(A) - 1 + \beta(A) \geq b + k.\tag{5.3}$$

For the canonical form there are $b + 1$ prefix vertices. Each of its $b - k$ nonzero integer vertices has one outgoing edge, and each of its k noninteger vertices has two. Hence

$$v(C(x)) = b + 1, \quad e(C(x)) = (b - k) + 2k = b + k, \quad \beta(C(x)) = k.\tag{5.4}$$

□

The cycle rank is the number of independent undirected cycles, not the number of simple cycles and not the directed cycle count. Directed cycles never occur. Thus the dyadic denominator has an exact representation-theoretic meaning: its exponent is the least undirected cycle rank achievable by any finite form.

Corollary 5.3. *A finite surreal value has an acyclic underlying representation if and only if it is an integer.*

Proof. Integers have canonical path representations. A noninteger has $k \geq 1$, so every representation has positive cycle rank by [theorem 5.2](#). □

The next lemma reproves [Corollary 5.3](#) by a direct edge count that never mentions prefixes, and it buys two things the first proof does not give: it identifies the acyclic graph as a *path*, and it isolates the unary evaluation identities (5.5), which are used repeatedly in [sections 7](#) and [9](#).

Lemma 5.4 (Forest obstruction). *If A is finite and $U(A)$ is a forest, then $U(A)$ is a path and $\text{val}(A)$ is an integer. Moreover, for a form Y of integer value n ,*

$$\text{val}(\{Y \mid \}) = \max(0, n + 1), \quad \text{val}(\{ \mid Y \}) = \min(0, n - 1).\tag{5.5}$$

Proof. The graph is connected, so a forest here is a tree: with v vertices it has $v - 1$ edges. Every vertex except the unique sink O has at least one outgoing edge, so the outdegree sum is at least $v - 1$; equality forces every other than O to have exactly one option. Since the root reaches every vertex, the whole graph is a single directed path ending at O .

For (5.5), if $n < 0$ then zero satisfies the cut $\{Y \mid \}$ and is simplest; if $n \geq 0$ the simplest number strictly above n is $n + 1$. The right-hand identity is the reflection. Starting from $\text{val}(O) = 0$ and proceeding up the path, every value is an integer. □

5.2 Classifying every vertex-minimal form

Theorem 5.5 (The vertex-minimal forms). *Fix $x \in \mathbb{D}$, with prefixes $p_0, \dots, p_b$, and put $k = \kappa(x)$. A finite form A of value x has exactly $b + 1$ vertices if and only if it can be built as follows. Use one vertex A_j of value p_j for each $0 \leq j \leq b$, with $A_0 = O$. At each A_j , include the canonical nearest-prefix options. Independently add or omit any other earlier-prefix option A_i with $i < j$, placing it on the left when $p_i < p_j$ and on the right when $p_i > p_j$.*

Consequently every vertex-minimal form contains $D(C(x))$ as a rooted, value- and side-preserving spanning subgraph, and

$$\#\{A : \text{val}(A) = x, \text{v}(A) = b + 1\} = 2^{\binom{b+1}{2} - (b+k)} = 2^{b(b-1)/2 - k}. \quad (5.6)$$

Proof. Suppose first that A has $b + 1$ vertices. The path in [theorem 4.6](#) already contains at least this many vertices. Therefore it exhausts the graph and has no extra vertices between the prefixes. Denote its vertices by $A_b, A_{b-1}, \dots, A_0$, where $\text{val}(A_j) = p_j$. Every edge must go from some A_j to an earlier A_i : an edge in the reverse direction would combine with the path to create a directed cycle.

Fix $j > 0$. If there is an earlier prefix below p_j , let ℓ be the largest such value. The vertex of value ℓ must be a left option of A_j . Otherwise all actual left-option values are strictly below ℓ , while all right-option values are strictly above p_j and hence above ℓ . The simpler value ℓ would then satisfy the cut of A_j , a contradiction. The smallest earlier prefix above p_j , when present, is likewise a mandatory right option. By [Lemma 4.4](#), these are precisely the canonical options. Every other possible edge has its side determined by numerical comparison.

Conversely, perform the stated construction in increasing order of j . Inductively, all earlier vertices have their assigned prefix values. The mandatory nearest options already give the cut value p_j . Adding more earlier prefixes on their permitted sides changes neither extreme bound, so it does not change the value. All vertices are distinct because their values are distinct. The consecutive-prefix edges are mandatory, so the root reaches every vertex and the closure has exactly $b + 1$ vertices.

There are $\binom{b+1}{2}$ possible edges from later to earlier prefixes, each with a uniquely determined side. Of these, exactly $b + k$ are mandatory by [theorem 5.2](#); every remaining edge is independently optional. Distinct choices produce distinct forms: from the final form one recovers its uniquely valued prefix vertices and all their edge choices. This proves the count and the spanning statement. $\square$

Notice what the theorem does *not* say. The vertex A_j need not be literally identical to $C(p_j)$, because it may have extra options. The spanning embedding sends the canonical vertex of value p_j to A_j .

Corollary 5.6 (Unique edge-minimizer). *The canonical form $C(x)$ is the unique finite form of value x with $e = b + k$.*

Proof. Equality in the edge bound forces both $v = b + 1$ and $\beta = k$. The classification then permits no optional edges, because its mandatory edges already number $b + k$. The recursive construction is therefore exactly (2.5). $\square$

Value x	b	k	Minimum vertices	Minimum edges	Vertex-minimal forms
0	0	0	1	0	1
1/2	2	1	3	3	1
1/4	3	2	4	5	2
3/4	3	2	4	5	2
3/2	3	1	4	4	4
3	3	0	4	3	8
5/8	4	3	5	7	8
5/2	4	1	5	5	32
4	4	0	5	4	64

Table 1: The exact minima and enumeration. Negating a value gives the same counts. The accompanying exhaustive computation checks all value classes with $b \leq 4$, including every row in this table.

6 Sharp smallness and the failure of minimum rank

6.1 Four vertices are never enough

Both halves of this section, and also the exact extremal function of [section 7](#), rest on one case analysis: the possible dependency orientations of a quadrilateral. We isolate it once.

Lemma 6.1 (Triangle-free forms on at most four vertices). *Let A be a finite form with at most four vertices whose underlying graph $U(A)$ is triangle-free. Then $\text{val}(A)$ is an integer. Equivalently, every finite form of noninteger value on at most four vertices has a triangle in its underlying graph.*

Proof. If $U(A)$ has no cycle, [Lemma 5.4](#) gives an integer value. Otherwise $U(A)$ is connected, triangle-free, has at most four vertices, and has a cycle, so it is exactly C_4 .

The dependency orientation of this quadrilateral has a unique source, namely the root, and a unique sink, namely O . It therefore consists of two internally disjoint directed root-to- O paths whose lengths sum to four: either $(2, 2)$ or $(1, 3)$. All their intermediate vertices are unary forms, so [\(5.5\)](#) evaluates them.

In the $(2, 2)$ case the two intermediate forms are distinct unary forms over O , namely $C(-1)$ and $C(1)$, of values -1 and 1 . If the root has only one nonempty option side, its value is an integer by [Lemma 5.1](#). If both sides are nonempty, the only valid arrangement is $\{C(-1) \mid C(1)\}$, of value zero.

In the $(1, 3)$ case one root option is O itself, and the other is a unary form over a unary form over O , so by [\(5.5\)](#) its value lies in $\{-2, 0, 2\}$. A one-sided root again has integer value by [Lemma 5.1](#). If the two options lie on opposite sides, the value 0 is impossible, since a cut cannot have equal bounds on its two sides; the remaining intervals are $(-2, 0)$ and $(0, 2)$, whose simplest elements are -1 and 1 . Reversing which path is the long one is the same argument.

In every case the value is an integer. □

Theorem 6.2 (Smallest size and rank of a counterexample). *Every finite form with at most four vertices contains its canonical underlying graph as an unlabelled undirected subgraph. Every finite form of rank at most two has the same property. Hence the example in [theorem 3.1](#) is smallest both in vertex count and in form rank.*

Proof. Let A have $N \leq 4$ vertices and value x . By [theorem 4.6](#), $b(x) \leq N - 1 \leq 3$. If $N = b(x) + 1$, the spanning theorem [5.5](#) applies. If $x = n$ is an integer, the canonical graph is a path of length $|n|$. The prefix-path theorem supplies a directed path of at least that length in $D(A)$, so its undirected version contains the required path.

The only remaining possibility is a noninteger of birthday at most two with $N = 4$: thus $x = 1/2$ or $x = -1/2$. Its canonical graph is a triangle. By [Lemma 6.1](#) a four-vertex form with a triangle-free graph has integer value, so $U(A)$ has a triangle and the required embedding exists.

Finally, a form of rank at most two has no proper subforms other than O , $C(-1)$, and $C(1)$. Its closure therefore has at most four vertices, so the first part applies. The pentagon has five vertices and rank three. □

Note which hypotheses each half uses. [Lemma 6.1](#) is purely local: it inspects four vertices and never mentions prefixes, and it is what supplies the lower bound $m_{1/2}(4) \geq 5$ in [theorem 7.7](#). The rank half of [theorem 6.2](#) and the four-vertex cases with $N = b(x) + 1$ instead go through the vertex-minimal spanning theorem, which [Lemma 6.1](#) alone cannot reach: a four-vertex form of value $1/4$ has a triangle-containing graph, so the lemma says nothing, whereas [theorem 5.5](#) shows that its graph is the canonical diamond.

This proof is independent of the exhaustive search. The computation in [section 12](#) supplies a separate audit of the small cases and produces all sixteen counterexamples at five vertices.

6.2 Least birthday is not enough

It is tempting to rescue containment by requiring the representation to be born as early as its value. That weaker size restriction does not work.

Proposition 6.3 (A rank-minimal non-containing form). *Let*

$$O = \{|\}, \quad P = \{O|\}, \quad Z' = \{|\ P\}, \quad H = \{O|P\}, \quad Q = \{Z'|H\}. \quad (6.1)$$

Then $\text{val}(Q) = 1/4$ and $\rho(Q) = b(1/4) = 3$, but $U(C(1/4))$ is not a subgraph of $U(Q)$.

Proof. The values of Z' and H are zero and $1/2$, so the value of Q is $1/4$. Their ranks are both two, giving $\rho(Q) = 3$. The canonical quarter graph has vertices $C(1/4), H, P, O$ and edges

$$C(1/4)O, \quad C(1/4)H, \quad HO, \quad HP, \quad PO. \quad (6.2)$$

It is the diamond K_4 with one edge removed, and contains two distinct triangles. The graph of Q has edge set

$$QZ', \quad QH, \quad Z'P, \quad HO, \quad HP, \quad PO. \quad (6.3)$$

Its only triangle is HPO ; the remaining cycle is the quadrilateral $QZ'PHQ$. An injective subgraph map would send the two distinct triangles of the diamond to two distinct triangles, which is impossible. $\square$

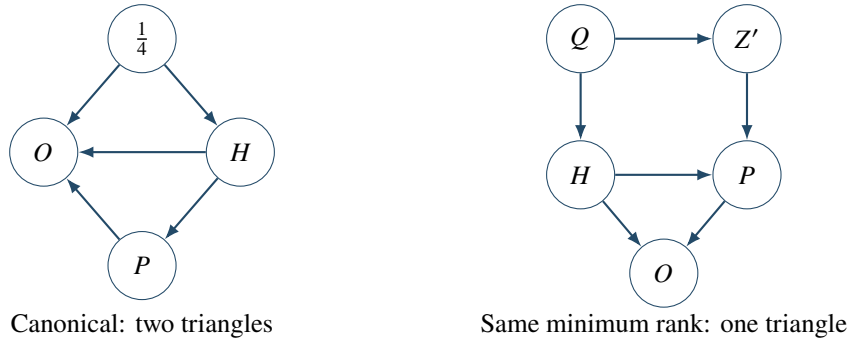


Figure 3: Rank minimality does not force containment. Both roots have value $1/4$ and rank three. Option-side labels are omitted; the defining cuts are (2.5) and (6.1).

Thus vertex minimality, rank minimality, and edge minimality are genuinely different conditions. Vertex minimality permits many forms but forces the canonical edges. Edge minimality forces the canonical form itself. Rank minimality alone permits a counterexample.

Remark 6.4 (A question answered). [Proposition 6.3](#) settles, negatively, a question that the sparse side of this investigation had left open: must every form whose rank equals the intrinsic birthday of its value contain the canonical option graph of that value? The pentagons of [section 3](#) do not settle it, because $\rho(F) = 3$ and $\rho(G) = 4$ while $b(1/2) = 2$, so neither is rank-minimal. The form Q is rank-minimal and still fails containment even under the unlabelled convention, so it also rules out any stronger subgraph requirement preserving roots, directions, option sides, or values. Questions about minors or restricted arithmetic presentations are different problems; see [Questions 13.2](#) and [13.5](#). The transfinite example of [Proposition 11.2](#) shows that at a limit birthday the analogous unrestricted statement can fail for a different reason.

7 Cycle representations of one half, and exact optimality

Each of the two pentagons extends to its own infinite family of cycles, and a third family supplies the even lengths. Together they realize every cycle length except four, and this is enough to determine the exact girth–size tradeoff at the value $1/2$.

7.1 First odd family: a zero spine through -1

The rank-minimal pentagon extends to a particularly transparent infinite family. Define a zero-valued spine by

$$Z_0 = O, \quad N_i = \{ \mid Z_i \}, \quad Z_{i+1} = \{ N_i \mid \} \quad (i \geq 0). \quad (7.1)$$

Lemma 7.1. *For every $i \geq 0$,*

$$\text{val}(Z_i) = 0, \quad \rho(Z_i) = 2i, \quad \text{val}(N_i) = -1, \quad \rho(N_i) = 2i + 1. \quad (7.2)$$

The graph of Z_i is a path of length $2i$, with alternating values $0, -1$. All vertices on this path are structurally distinct.

Proof. Induct on i . A right-only cut bounded by zero has value -1 ; a left-only cut bounded by -1 has value zero. Each construction raises rank by one. Distinct ranks give distinct forms, and every noninitial spine vertex has exactly one option, namely the preceding vertex of the path. $\square$

Theorem 7.2 (Odd-cycle family). *Let $P = C(1)$ and, for $m \geq 1$, set*

$$F_m = \{ Z_m \mid P \}. \quad (7.3)$$

Then

$$\text{val}(F_m) = \frac{1}{2}, \quad \rho(F_m) = 2m + 1, \quad U(F_m) = C_{2m+3}, \quad \beta(F_m) = 1. \quad (7.4)$$

Proof. The two root-option values are zero and one. The left root-to- O path has length $2m + 1$, and the right path $F_m \rightarrow P \rightarrow O$ has length two. Their internal vertices are disjoint, since the spine values are only zero and -1 , whereas P has value one. The two paths therefore form a chordless cycle of length $2m + 3$. The rank is one more than that of Z_m , and a cycle has cycle rank one. $\square$

For $m = 1$ this is $F_1 = \{ Z_1 \mid P \}$, of rank three; it is the pentagon of (1.1), since $Z_1 = \{ N_0 \mid \}$ and $N_0 = \{ \mid O \}$.

This family shows that the least possible cycle rank does not imply a bound on girth, even for the single fixed value $1/2$. It also supplies counterexamples whose distance from the canonical local triangle can be made arbitrarily large in the ordinary graph-theoretic sense of excluding every short cycle.

7.2 Second odd family: an alternating 0/1 spine

The second witness generalizes along a different spine, on the other side of the root, and through nonnegative values only. What this family buys is that it contains the canonical triangle itself as its initial member, so the passage from the canonical form to a counterexample happens inside one uniform construction. Put $K_0 = O$ and

$$K_{j+1} = \begin{cases} \{ K_j \mid \}, & j \text{ even,} \\ \{ \mid K_j \}, & j \text{ odd.} \end{cases} \quad (7.5)$$

Lemma 7.3. *For every $j \geq 0$ we have $\rho(K_j) = j$, and $\text{val}(K_{2r}) = 0$, $\text{val}(K_{2r+1}) = 1$. The forms $K_0, \dots, K_j$ are pairwise distinct and span a path of length j .*

Proof. Each step is a unary cut, so the rank increases by one and the new form has the previous one as its only option; distinct ranks give distinct forms. The values follow from (5.5): from value 0 a left unary cut gives 1, and from value 1 a right unary cut gives 0. $\square$

Theorem 7.4 (Second odd-cycle family). *For $r \geq 0$ put $G_r = \{ K_0 \mid K_{2r+1} \}$. Then*

$$\text{val}(G_r) = \frac{1}{2}, \quad \rho(G_r) = 2r + 2, \quad U(G_r) = C_{2r+3}, \quad \beta(G_r) = 1. \quad (7.6)$$

Here $G_0 = C(1/2)$ is the canonical triangle and $G_1 = G$ is the second witness (1.3).

Proof. The root options have values 0 and 1 by Lemma 7.3, so the cut is valid and its value is $1/2$. Its graph is the path $K_{2r+1} \rightarrow \dots \rightarrow K_0$ of length $2r + 1$ together with the two root edges to its endpoints, giving a chordless cycle of length $2r + 3$; no chord exists because each K_{j+1} has K_j as its only option and the root's only options are K_0 and K_{2r+1} . The rank is one more than $\rho(K_{2r+1}) = 2r + 1$, and a single cycle has cycle rank one. For $r = 0$ the form is $\{O \mid C(1)\}$, which is (2.5) at $1/2$; for $r = 1$ it is $\{O \mid K_3\}$ with $K_3 = P'$, which is (1.3). $\square$

The two odd families are genuinely different constructions and neither subsumes the other: they use different spine values (0, -1 against 0, 1), attach the long arm to opposite sides of the root, and have different ranks at the same cycle length, namely $2m + 1$ against $2r + 2$ for $C_{2m+3} = C_{2r+3}$. The first is the rank-economical one; the second passes through the canonical form.

7.3 An even family, and the full cycle spectrum

Neither odd family gives an even cycle. A third construction does. Recall $N_0 = \{ \mid O \}$, of value -1 , from (7.1), and set

$$N' = \{ \mid N_0 \}, \quad V_0 = \{ N' \mid \}, \quad V_{t+1} = \{ \mid \{ V_t \mid \} \} \quad (t \geq 0). \quad (7.7)$$

Theorem 7.5 (Even-cycle family). *For every $t \geq 0$ the form V_t has value 0 and rank $3 + 2t$, and the form*

$$G_t^{\text{even}} = \{ V_t \mid P \}, \quad P = C(1), \quad (7.8)$$

satisfies

$$\text{val}(G_t^{\text{even}}) = \frac{1}{2}, \quad \rho(G_t^{\text{even}}) = 4 + 2t, \quad U(G_t^{\text{even}}) = C_{6+2t}, \quad \beta(G_t^{\text{even}}) = 1. \quad (7.9)$$

Proof. By (5.5), $\text{val}(N') = -2$ and $\text{val}(V_0) = 0$, with ranks 2 and 3. Inductively, $\{ V_t \mid \}$ has value 1 and rank $4 + 2t$, so V_{t+1} has value 0 and rank $5 + 2t = 3 + 2(t + 1)$. All the forms on this branch are unary above N' , hence pairwise distinct by rank, and the branch from V_t down to O is a path with $3 + 2t$ edges.

The root cut has options of values 0 and 1, so $\text{val}(G_t^{\text{even}}) = 1/2$. The long branch and the short branch $P \rightarrow O$ meet only at O : the first form above O on the long branch is N_0 , of value -1 , not P , and every later form on that branch has rank at least two while $\rho(P) = 1$. The two internally disjoint root-to- O paths, of lengths $3 + 2t + 1$ and 2, therefore form a chordless cycle of length $6 + 2t$. $\square$

Proposition 7.6 (All cycle lengths except four). *For every integer $s = 3$ or $s \geq 5$ there is a finite form of value $1/2$ whose underlying option graph is exactly C_s . The length $s = 4$ is the unique unattainable one: no form of value $1/2$, or indeed of any noninteger value, has underlying graph C_4 .*

Proof. Odd lengths $s = 2r + 3 \geq 3$ come from theorem 7.4 and even lengths $s = 6 + 2t \geq 6$ from theorem 7.5. A form whose graph is C_4 has four vertices and no triangle, so Lemma 6.1 makes its value an integer. (Integers do realize C_4 : the form $\{C(-1) \mid C(1)\}$ has value zero and underlying graph C_4 . The exclusion is of noninteger values, not of the graph itself.) $\square$

7.4 The exact girth-size tradeoff at one half

Theorem 7.7 (Exact extremal function for $1/2$). *For an integer $g \geq 3$ let*

$$m_{1/2}(g) = \min \{ v(A) : \text{val}(A) = \frac{1}{2}, \text{girth}(U(A)) \geq g \}. \quad (7.10)$$

Then $m_{1/2}$ is given by (1.9): it equals 3 at $g = 3$, 5 at $g = 4$, and g for every $g \geq 5$. Moreover five vertices is the minimum over all triangle-free finite representations of all noninteger values, and $1/2$ attains that minimum.

Proof. By [theorem 5.2](#) every representation of $1/2$ has $\beta \geq 1$, so it contains a cycle; if its girth is at least g that cycle already uses at least g vertices, giving $m_{1/2}(g) \geq g$ for every g . For $g = 3$ the canonical triangle attains three. For $g \geq 5$, [Proposition 7.6](#) supplies a form whose whole graph is C_g , attaining g . For $g = 4$ the bound $m_{1/2}(4) \geq 4$ must be improved: by [Lemma 6.1](#) a four-vertex triangle-free form has integer value, so $m_{1/2}(4) \geq 5$, and either pentagon of [section 3](#) attains five. The same lemma applies verbatim to every noninteger value, and the half supplies a five-vertex triangle-free example. $\square$

Remark 7.8 (A question answered). [theorem 7.7](#) settles, for $x = 1/2$, the optimal-girth-cost question raised in [Question 13.3](#): the least number of vertices at girth at least g is exactly g once $g \geq 5$, so the linear dependence of the general bound (8.11) on g is of the right order and has optimal constant one at this value. Nothing here determines the constant for other noninteger values.

Remark 7.9. Even cycles of length at least six occur in [theorem 7.5](#), so a bipartite option graph need not represent an integer. The obstruction in [Lemma 5.4](#) is the absence of *all* cycles, not of odd cycles.

8 Arbitrarily large girth for every finite-birthday value

The preceding construction treats one value. A uniform construction requires more care: simply replacing repeated vertices by “fresh copies” is illegal because identical forms are the same vertex. We first unfold the canonical graph into an occurrence tree and then use different-rank zero forms to prevent unwanted identifications.

8.1 The occurrence tree and its zero leaves

Let $x > 0$ be dyadic. Recursively unfold $D(C(x))$ by giving every occurrence of an option its own tree vertex, even when that form occurs elsewhere. This produces a finite rooted occurrence tree T_x . It is an auxiliary tree, not yet a form graph. Every leaf is an occurrence of O , and every internal vertex has positive numerical value. Write

$$I = \#\{\text{internal occurrences}\}, \quad L = \#\{\text{leaf occurrences}\}. \quad (8.1)$$

Every root-to-leaf path has length at most $b = b(x)$.

Enumerate the leaves as $0, 1, \dots, L - 1$ in any fixed order; a left-before-right traversal will also give planarity. Choose a positive integer M satisfying

$$2M > b. \quad (8.2)$$

Replace leaf occurrence i by the genuine form Z_{Mi} from (7.1). Rebuild all internal cuts, preserving their original left/right choices. Denote the resulting root form by $A_{x,M}$.

Theorem 8.1 (Spaced-spine construction). *For a positive dyadic x and a positive integer M satisfying (8.2), the form $A_{x,M}$ has value x . Its dependency graph is exactly the occurrence tree with its distinct leaves attached to the corresponding distinct vertices of the zero spine. If d_* is the depth in T_x of the last leaf, then*

$$v(A_{x,M}) = I + 2M(L - 1) + 1, \quad (8.3)$$

$$e(A_{x,M}) = I + L - 1 + 2M(L - 1), \quad (8.4)$$

$$\beta(A_{x,M}) = L - 1, \quad (8.5)$$

$$\rho(A_{x,M}) = 2M(L - 1) + d_*. \quad (8.6)$$

The graph is acyclic if $L = 1$; otherwise

$$\text{girth}(U(A_{x,M})) \geq 2M + 2. \quad (8.7)$$

In every leaf order the maximum degree is at most three. If the leaves are enumerated by a left-before-right traversal, the graph is also planar.

Proof. Every replacement leaf has value zero, the value of the leaf it replaces. Induction upward through the finite occurrence tree therefore gives the original value at every rebuilt internal vertex, including value x at the root. In particular, every rebuilt internal value is positive.

We now prove that no two distinct internal occurrences collapse to one form. For an internal occurrence v and a leaf i below it, let $d(v, i)$ denote the number of tree edges from v to i . The recursive definition of rank gives

$$\rho(v) = \max_{i \text{ below } v} (2Mi + d(v, i)). \quad (8.8)$$

Every distance in this formula is between one and b . Because $2M > b$, the term with the largest leaf index $t(v)$ uniquely dominates all the others:

$$\rho(v) = 2Mt(v) + d(v, t(v)). \quad (8.9)$$

If two internal occurrences have different largest leaf indices, their ranks lie in disjoint intervals of length at most b , so the ranks differ. If they have the same largest leaf index, both are ancestors of that one leaf and are therefore comparable in the tree. Distinct ancestors have different distances to the leaf, again giving different ranks. Thus all rebuilt internal vertices are distinct forms.

No rebuilt internal vertex can equal any spine vertex, because internal values are positive, whereas all spine values are zero or -1 . The spine vertices are mutually distinct by [Lemma 7.1](#). The chosen leaf attachment points Z_{Mi} are consequently distinct. This proves the asserted exact graph structure, including the absence of hidden identifications.

The tree has $I + L$ vertices and $I + L - 1$ edges. The spine from $Z_{M(L-1)}$ to O has $2M(L - 1) + 1$ vertices and $2M(L - 1)$ edges. Identifying the L leaves with their attachment points gives (8.3) and (8.4). Subtracting yields (8.5). Formula (8.8) at the root gives (8.6).

Both the occurrence tree and the spine are individually acyclic. Any cycle in their union must therefore include a nonempty spine segment. Choose a maximal such segment within the cycle. Its endpoints must be distinct attachment points: an interior spine vertex which is not an attachment has no edge leaving the spine. Two attachment points are at spine distance at least $2M$. The complementary part of the cycle has length at least two, since distinct leaves of the occurrence tree are not joined by a tree edge. This proves the girth bound. If there is only one leaf, there is only one attachment and no cycle.

Every internal tree vertex has at most two children and one parent. Each attachment has one tree parent and at most two spine neighbours; all other spine vertices have degree at most two. Thus the maximum degree is at most three. For planarity, draw the ordered occurrence tree above a horizontal line with its leaves on the line in their traversal order. Such a drawing is obtained recursively by placing child subtrees in disjoint horizontal intervals; unary vertices cause no difficulty. Draw the spine between consecutive leaves below the line, inserting its intermediate vertices along each segment. The chosen leaf order prevents crossings. $\square$

The rank-separation argument is the central technical precaution. It replaces an unjustified appeal to labelled copies by an actual construction in the hereditary syntax of surreal forms.

Corollary 8.2 (Unbounded girth at every fixed value). *For every finite-birthday surreal value x and every positive integer g , there exists a finite form A of value x whose underlying graph has no cycle of length less than g .*

Proof. For positive x , take M large enough that $2M > b(x)$ and $2M + 2 \geq g$. For negative x , construct a form for $-x$ and negate it recursively, exchanging left and right options. Negation is an involution on forms and induces an isomorphism of underlying graphs. For zero, use O . We regard acyclic graphs as having infinite girth. $\square$

Corollary 8.3 (Avoiding every fixed cyclic graph). *Fix a finite-birthday surreal value x and a finite graph H containing a cycle. Some finite representation of x has no subgraph isomorphic to H .*

Proof. Choose the representation in [Corollary 8.2](#) with girth greater than $|V(H)|$. Any copy of H would contain a cycle of length at most $|V(H)|$. $\square$

8.2 Size bounds for the construction

The construction is explicit, but unfolding can enlarge the canonical graph. The number of leaves has a simple Fibonacci bound. Let $f_1 = f_2 = 1$ and $f_{r+2} = f_{r+1} + f_r$.

Proposition 8.4 (A Fibonacci bound on the unfolded leaves). *If $x \in \mathbb{D}$ and $k = \kappa(x)$, the canonical occurrence tree has*

$$L \leq f_{k+2}. \quad (8.10)$$

For each $k \geq 0$, equality is attained by some value with denominator exponent k . Moreover, $I \leq b(x)L$.

Proof. Every integer has a unary occurrence tree and hence one leaf. In any unit interval, assign weight one to each integer endpoint. The number of leaves at a dyadic midpoint is the sum of the leaf counts of its two endpoint forms. Thus binary subdivision updates an endpoint-weight pair (a, c) to either $(a, a + c)$ or $(a + c, c)$, starting with $(1, 1)$.

After r updates, the larger endpoint weight is at most f_{r+2} and the smaller at most f_{r+1} . This follows by induction: the new larger weight is the sum of the old pair, and the retained weight is at most the old larger one. A value of reduced denominator 2^k is the midpoint after $k - 1$ updates, so its leaf count is at most f_{k+2} . At each update retain the larger of the previous two weights. The successive sorted pairs are consecutive Fibonacci numbers, proving attainability. The case $k = 0$ is the integer case.

Finally, every internal occurrence lies on a path to some leaf, and each root-to-leaf path contains at most $b(x)$ internal vertices. Counting internal-vertex/descendant-leaf incidences yields $I \leq b(x)L$. $\square$

The same bound has a second, shorter proof, which we record because it explains the Fibonacci recursion arithmetically rather than through the geometry of binary subdivision. It is stated for the binary cut tree $T(x)$ of [section 9](#), which is the tree obtained from T_x by stopping at integer leaves; the two trees have the same leaf count, because the unfolding below an integer leaf is a unary chain and contributes exactly one O -leaf.

Proposition 8.5 (A denominator-sensitive bound, second proof). *If x has reduced denominator 2^k with $k \geq 1$, then $L \leq f_{k+2}$.*

Proof. An integer contributes one leaf and a half-integer two, matching $f_2 = 1$ and $f_3 = 2$. Let $x = m/2^k$ with m odd and $k \geq 2$. Of the two numerators $m - 1$ and $m + 1$, exactly one has 2-adic valuation one and the other has valuation at least two, a zero numerator being read as an integer endpoint. Hence the two canonical options have reduced denominator exponents exactly $k - 1$ and at most $k - 2$. Their leaf counts add, so induction bounds the total by $f_{k+1} + f_k = f_{k+2}$. $\square$

This argument does not by itself establish attainment; the endpoint-weight recursion in [Proposition 8.4](#) does, for every k . The computations in [section 12](#) confirm attainment numerically for $k \leq 8$, with maximum leaf counts 2, 3, 5, 8, 13, 21, 34, 55, and separately over all denominator exponents 0 through 12.

Combining the proposition with (8.3) gives the concrete bound

$$v(A_{x,M}) \leq b(x)f_{k+2} + 2M(f_{k+2} - 1) + 1. \quad (8.11)$$

This is a graph-size bound, not a claim of optimality. The cycle rank produced by the construction is $L - 1$, which can exceed the minimum k . For example, the canonical unfolding at $5/8$ has five leaves, so the construction has cycle rank four, while the minimum possible cycle rank for that value is three.

x	M	L	I	ρ	v	e	β	Girth
$1/2$	4	2	2	10	11	11	1	11
$3/4$	4	3	4	18	21	22	2	11
$5/8$	4	5	7	35	40	43	4	11
$5/16$	4	8	10	60	67	73	7	11
$7/4$	4	3	7	19	24	25	2	13
$-5/8$	4	5	7	35	40	43	4	11

Table 2: Exact outputs of the spaced-spine implementation. The proved lower bound for $M = 4$ is girth at least ten; the actual girths in these examples are larger. Leaves are enumerated by a left-before-right recursive traversal.

9 A planar, degree-three construction: tree and spine

[theorem 8.1](#) already gives arbitrarily large girth at every finite-birthday value, with exact vertex, edge, cycle-rank, and rank formulas. Its ordered-leaf version is planar and has maximum degree at most three. This section gives a second construction with those graph properties: it stops the canonical tree at integer leaves, attaches them to an integer-valued spine, and uses a different proof of non-collapse. Its spacing requirement depends only on the requested girth, whereas the first construction requires $2M > b(x)$. The resulting size and rank bounds are expressed in the integer leaf labels and the denominator exponent.

The contrast between the two non-collapse proofs is the heart of the matter. [theorem 8.1](#) separates rebuilt vertices by *rank arithmetic*: the spacing condition $2M > b$ makes $\rho(v) = 2Mt(v) + d(v, t(v))$ single out a dominant leaf. The construction below separates them by *integrality* — internal vertices are nonintegers, spine vertices are integers — followed by a descent through a singleton-option template. Each proof must establish that distinct occurrences remain distinct forms: without that step, the graph argument would describe allocated objects rather than mathematical forms.

9.1 Unfolding the canonical cut down to integers

Let x be a noninteger dyadic with $\kappa(x) = k$. Starting from (2.5), unfold every occurrence of a noninteger into its two canonical options, but stop as soon as an integer is reached, and do not identify repeated occurrences. The result is a finite ordered full binary tree $T(x)$.

Every internal vertex of $T(x)$ carries a noninteger dyadic label and every leaf an integer label. If $x \in (n, n+1)$, every leaf label lies in $\{n, n+1\}$, since the canonical options of a noninteger in $(n, n+1)$ stay in $[n, n+1]$. The reduced denominator exponent strictly decreases along every internal edge, so the tree is finite and its depth is at most k . Write $\ell = \ell(x)$ for the number of leaves and enumerate them from left to right with integer labels $n_1, \dots, n_\ell$. A full binary tree with ℓ leaves has $\ell - 1$ internal vertices and $2\ell - 2$ edges. For example,

$$\frac{5}{8} = \left\{ \frac{1}{2} \mid \frac{3}{4} \right\}, \quad \frac{3}{4} = \left\{ \frac{1}{2} \mid 1 \right\}, \quad \frac{1}{2} = \{0 \mid 1\} \quad (9.1)$$

unfolds to a tree with five leaves; at this stage the two occurrences of $1/2$ are separate subtrees. The leaf count obeys the same Fibonacci bound as before, by [Proposition 8.4](#) or [Proposition 8.5](#).

9.2 A unary spine with separated integer markers

Lemma 9.1 (Separated integer markers). *Given a nonempty finite list of integers $n_1, \dots, n_\ell$ and an integer $d \geq 1$, there is a unary chain of pairwise distinct forms*

$$S_0 = O, \quad S_1, \dots, S_N, \quad (9.2)$$

each S_{i+1} having S_i as its unique option, and marked indices $0 < s_1 < \dots < s_\ell = N$ such that

$$\text{val}(S_{s_j}) = n_j, \quad s_1 \geq d, \quad s_{j+1} - s_j \geq d. \quad (9.3)$$

With $s_0 = 0$ it can be arranged that

$$s_j - s_{j-1} \leq d + |n_j| + 2. \quad (9.4)$$

Every spine form has rank equal to its index.

Proof. Each stage begins at the last constructed spine form, whose value is an integer. If that value is positive, apply a unary right cut; if negative, a unary left cut; (5.5) shows that either resets the value to zero, and no step is needed if the value is already zero.

From a zero-valued form, the two unary steps

$$Y \mapsto \{Y \mid \} \mapsto \{ \mid \{Y \mid \} \} \quad (9.5)$$

return to value zero through value one, again by (5.5). Repeat this pair $\lceil d/2 \rceil$ times, which inserts between d and $d + 1$ edges. Finally take $|n_j|$ unary left steps from the resulting zero when $n_j > 0$, or $|n_j|$ unary right steps when $n_j < 0$; the value reaches n_j . Mark that index.

The stage uses at least d edges and at most $1 + (d + 1) + |n_j|$, which is (9.4). Each new form has the previous one as its only option, so its rank is one larger; distinct ranks make all spine forms distinct. $\square$

9.3 Substituting the markers

Replace the j th integer leaf occurrence of $T(x)$ by the actual form S_{s_j} , and rebuild every internal vertex as the singleton-left, singleton-right cut with the same tree pattern as before. Call the resulting root $X_{x,d}$.

Lemma 9.2 (Value preservation). *Every rebuilt internal vertex is a valid numerical form whose value is its original label in $T(x)$. In particular $\text{val}(X_{x,d}) = x$.*

Proof. At the leaves the assertion is Lemma 9.1. At an internal vertex the two rebuilt option values are exactly the two values of the corresponding canonical cut, so they are strictly ordered and the simplest number between them is the original label. Bottom-up induction gives validity and preserves every value. $\square$

Lemma 9.3 (No unintended identifications). *Distinct internal occurrences of $T(x)$ become distinct forms; none equals a spine form; and distinct leaf occurrences are attached to distinct spine forms.*

Proof. Spine forms have integer values, while by Lemma 9.2 every rebuilt internal form has a noninteger value; so the two classes are disjoint. The markers are distinct by Lemma 9.1.

Suppose two different internal occurrences became the same form. They would then have the same value, say r , and the unfolded canonical template below an occurrence labelled r is determined by r alone. These occurrences cannot be ancestors of one another, since the denominator exponent strictly decreases along internal edges. Their descendant subtrees are therefore disjoint. Since each rebuilt left and each rebuilt right option set is a singleton, equality of the two forms forces equality of their left children, then of their right children, and inductively of their corresponding descendants along every fixed left/right path in that template. Follow one such path down to an integer leaf. The two occurrences then force two corresponding markers to coincide. But every leaf occurrence of the whole unfolded tree received a distinct marker, a contradiction. $\square$

The lemma establishes the exact graph, not merely an upper bound for it: $U(X_{x,d})$ consists of the spine path together with the binary tree $T(x)$, whose ℓ leaves are attached to distinct marked spine vertices. No further merging and no further edge is present.

9.4 Girth, degree, and planarity

Lemma 9.4 (Graph bounds). *The graph $U(X_{x,d})$ is planar, has maximum degree at most three, and has girth at least $d + 2$.*

Proof. Both the tree and the spine are acyclic, so a simple cycle in their union uses edges of both kinds. On such a cycle take a maximal nonempty segment of spine edges. Its endpoints are two different attachment vertices, because only those have edges leaving the spine, and their indices differ by at least d ; so the segment has at least d edges. The cycle also uses at least two tree edges, one to leave the spine and one to return. Its length is therefore at least $d + 2$.

For the degrees: an internal non-root tree vertex has two children and one parent; the root has two children only; a marked spine vertex has at most two spine neighbours and one tree parent; an unmarked spine vertex has at most two neighbours. So every degree is at most three.

For planarity, draw the ordered binary tree with its leaves in left-to-right order on a horizontal line and all other tree vertices above that line; such a drawing exists recursively, by placing the two subtrees in disjoint horizontal intervals. Join consecutive marked leaves by a path drawn below the line, inserting the required unmarked spine vertices between them, and extend the initial segment down to S_0 to the left. No crossings are introduced. The markers were assigned in exactly this leaf order, so the drawing represents the actual graph. $\square$

Theorem 9.5 (Planar high-girth representation theorem). *For every dyadic x and every integer $g \geq 3$ there is a finite numerical form X with $\text{val}(X) = x$ whose underlying option graph is planar, has maximum degree at most three, and has girth at least g . A forest is assigned infinite girth.*

Proof. If x is an integer, take the canonical path $C(x)$, which is acyclic and has maximum degree at most two. Otherwise apply the construction with $d = g - 2$ and use [Lemmas 9.2 to 9.4](#). $\square$

9.5 Explicit size and height bounds

Let $\ell = \ell(x)$, let $k = \kappa(x)$, and put $\mu = \max_j |n_j|$. Summing (9.4) gives $N \leq \ell(d + \mu + 2)$. There are $N + 1$ spine vertices and $\ell - 1$ new internal vertices, so

$$v(X_{x,d}) = N + \ell \leq \ell(d + \mu + 3), \quad (9.6)$$

and, since a directed path traverses at most k internal tree edges before entering the spine and then at most N spine edges,

$$\rho(X_{x,d}) \leq N + k \leq \ell(d + \mu + 2) + k. \quad (9.7)$$

The edge count is exact:

$$e(X_{x,d}) = N + 2\ell - 2, \quad \beta(X_{x,d}) = e - v + 1 = \ell - 1. \quad (9.8)$$

Raising the required girth therefore lengthens the spine without increasing the number of independent cycles. For a fixed x both bounds are linear in the requested girth.

These bounds are stated in ℓ , μ , and d , whereas (8.3) and (8.11) are stated in I , L , $b(x)$, and M . The cycle rank $\ell - 1$ here equals $L - 1$ there: extending each integer leaf by its canonical unary chain to zero does not change the leaf count. Neither construction is claimed to be minimal. For the single value $1/2$ the cycle constructions of [section 7](#) are far smaller, and [theorem 7.7](#) shows they are optimal.

10 Which canonical graphs are unavoidable?

The large-girth theorems answer much more than the single half-valued example. We give the classification twice, under two containment conventions of different strength, with two independent proofs of the positive half. Both negative halves reduce to the same observation, which we isolate first.

Lemma 10.1 (A triangle in every noninteger canonical graph). *If $x \in \mathbb{D} \setminus \mathbb{Z}$ then $U(C(x))$ contains a triangle.*

Proof. Suppose first that $\kappa(x) = 1$, so $x = n + \frac{1}{2}$ for an integer n . The two canonical options are $C(n)$ and $C(n+1)$, and these adjacent integer forms are joined by an edge: $C(n+1) \rightarrow C(n)$ when $n \geq 0$, and $C(n) \rightarrow C(n+1)$ when $n \leq -1$. With the root they form a triangle.

If $\kappa(x) = k > 1$, one canonical option has denominator exponent exactly $k-1$ by Remark 4.5. Following such options reaches a half-integer, whose complete option graph occurs inside $D(C(x))$, and the previous paragraph supplies a triangle there. $\square$

10.1 The unlabelled classification

Theorem 10.2 (Exactly the integers, unlabelled version). *For $x \in \mathbb{D}$, the graph $U(C(x))$ is an unlabelled undirected subgraph of every finite representation of x if and only if $x \in \mathbb{Z}$.*

Proof. If $x = n$ is an integer, $U(C(n))$ is a path of length $|n|$. The path provided by theorem 4.6 has at least that many edges. Select a subpath of the required length.

If x is nonintegral, $U(C(x))$ contains a triangle by Lemma 10.1. By Corollary 8.2, or equally by theorem 9.5 with $g = 4$, the value x has a representation with no triangle at all. Such a representation cannot contain $U(C(x))$. $\square$

The positive direction here uses unlabelled paths. It does not claim that consecutive integer values always occur on consecutive option edges, or that the integer classification extends unchanged to more restrictive kinds of embedding. The negative direction is stronger: it already rules out the weakest unlabelled notion.

10.2 The rooted-colour classification

The disclaimer just made is not the last word: the positive half does survive a considerably stronger convention, by a different argument. Say that the canonical option graph is *rooted-colour unavoidable at value x* if every finite form of value x admits an injective embedding of $D(C(x))$ which preserves the root, the arrow directions, and the L/R edge labels. Numerical vertex labels are ignored, the subgraph need not be induced, and the image of the canonical zero vertex is not required to be the sink of the ambient graph. This demand is strictly stronger than the one in theorem 10.2, which forgets all four of root, direction, side, and value.

Theorem 10.3 (Exactly the integers, rooted-colour version). *For $x \in \mathbb{D}$, the graph $D(C(x))$ is rooted-colour unavoidable at value x if and only if $x \in \mathbb{Z}$. Moreover, for every noninteger x there is a finite representation of x that excludes $U(C(x))$ even as an unrooted, unlabelled, undirected subgraph.*

Proof. Let $x = n > 0$ and let A be any finite form of value n . For a nonnegative integer a , comparison with $C(a)$, which has no right options, gives

$$\text{val}(B) > a \implies \text{some left option of } B \text{ has value at least } a. \quad (10.1)$$

Indeed, if no left option of B had value at least a , then (3.2) would give $B \leq C(a)$, that is $\text{val}(B) \leq a$.

Starting at the root, with value n , apply (10.1) successively at the thresholds $n - 1, n - 2, \dots, 0$. This produces a directed path of n left-option edges from the root. Ranks strictly decrease along option edges, so its vertices are distinct. The canonical graph of n is exactly such a path of L -edges rooted at its top, and mapping its successive vertices onto the constructed path gives the required root-, direction-, and side-preserving embedding. Reflection gives a path of right-option edges for a negative integer; for zero the canonical graph is a single vertex.

If x is noninteger, theorem 9.5 with $g = 4$ supplies a triangle-free representation, while $U(C(x))$ has a triangle by Lemma 10.1. So even the weakest containment fails, and a fortiori the rooted-colour one does. $\square$

Remark 10.4 (Scope of the two positive halves). The two proofs are independent and neither subsumes the other's method. theorem 10.2 routes through prefix occurrence, a simplicity-theoretic fact about sign expansions; theorem 10.3 routes through the recursive comparison (3.2) and never mentions sign expansions. What the second buys is the stronger conclusion: the embedded path is rooted at the actual root and consists of genuine left-option edges in the correct order. What neither asserts is that the embedded vertices carry the canonical integer values, or that the canonical form occurs as a literal subform. The negative halves have no such sensitivity: absence of a triangle rules out every edge-preserving interpretation at once.

11 What happens at a transfinite birthday?

The finite results already settle the selected question. The prefix argument, however, also yields a useful observation about ordinal-indexed simplicity. For an arbitrary set-sized well-founded form define

$$\rho(A) = \sup\{\rho(B) + 1 : B \in L_A \cup R_A\}. \quad (11.1)$$

The supremum of the empty set is zero. At limit ranks this formula must not be replaced by taking the supremum of the option ranks and then adding one.

theorem 4.3 was already proved for this ordinal rank. Thus every sign prefix of a value occurs somewhere in every well-founded representation. What can fail at a limit is the existence of one globally coherent choice of representatives for all prefixes. The finite concatenation in theorem 4.6 does not justify a transfinite concatenation.

Proposition 11.1 (A limit-stage coherence obstruction). *There is a well-founded form of value ω and rank ω in which every positive integer occurs as a subform value, but it is impossible to choose vertices B_n of value n , for all $n \geq 1$, such that B_{n+1} reaches B_n for every n .*

Proof. Use the zero spine (7.1). For $n \geq 1$, define

$$A_{n,0} = Z_n, \quad A_{n,j+1} = \{A_{n,j} \mid\} \quad (0 \leq j < n), \quad (11.2)$$

and then put

$$\Omega_* = \{A_{n,n} : n \geq 1 \mid\}. \quad (11.3)$$

Induction gives

$$\text{val}(A_{n,j}) = j, \quad \rho(A_{n,j}) = 2n + j. \quad (11.4)$$

The root cut is numerically the cut above all positive integers with no right bound, so it has value ω . Its rank is

$$\rho(\Omega_*) = \sup_{n \geq 1} (3n + 1) = \omega. \quad (11.5)$$

For $j > 0$, positive vertices in different arms are distinct: equality of values would force the same j , and equality of ranks would then force the same n . A positive vertex in arm n has a single outgoing edge to the

preceding vertex of that same arm. Once the path reaches its zero spine, it encounters only zero and -1 , never a positive vertex in another arm. Thus positive vertices in distinct arms cannot reach one another.

Every vertex of value one belongs to a particular finite arm, say arm n_0 . The only positive finite-valued ancestors of that vertex are in the same arm and have value at most n_0 . A coherent sequence B_n would make every B_n reach B_1 , including values larger than n_0 , which is impossible. Each individual integer nevertheless occurs in all sufficiently long arms. $\square$

There is no conflict with well-foundedness. Every descending option path from Ω_* first chooses a finite arm and then has finite length. The ordinal rank is ω because these finite lengths are unbounded, not because one infinite descending option path exists. Nor does the proposition contradict prefix occurrence: it separates individual occurrence from a simultaneous reachability-compatible selection.

11.1 Large girth at a limit birthday costs nothing

The second transfinite result concerns a different phenomenon and is logically independent of [Proposition 11.1](#): the first says that coherent prefix selection can fail at a limit, the second says that sparsity can be achieved at a limit without paying any birthday. Neither is used in the proof of the other.

Extend the canonical notation by

$$C(\omega) = \{C(n) : n \in \mathbb{N} \mid \}, \quad (11.6)$$

whose underlying graph is the infinite path of nonnegative integer forms together with a root joined to every vertex; it contains triangles. For an integer $d \geq 1$ define instead

$$W_d = \{C(dn) : n \in \mathbb{N} \mid \}. \quad (11.7)$$

The options are a cofinal subsequence of the canonical integer forms, not new copies of them.

Proposition 11.2 (Exact high-girth forms of ω). *For each $d \geq 1$ the form W_d has value ω and rank ω . Its underlying undirected option graph has girth exactly $d + 2$, and every vertex other than the root has degree at most three.*

Proof. Every left option of W_d is finite, so comparison with ω , which has no right options, gives $W_d \leq \omega$ by (3.2). Conversely W_d exceeds each of its options $C(dn)$, and those are unbounded in $\mathbb{N}$, so every finite integer lies below W_d ; applying (3.2) in the other direction gives $\omega \leq W_d$. Hence $\text{val}(W_d) = \omega$.

For rank, (11.1) gives $\rho(W_d) = \sup_{n < \omega} (dn + 1) = \omega$. Because the options are unbounded, their descendants comprise every canonical finite integer form, so the underlying graph is precisely the integer path together with a root joined at the vertices $0, d, 2d, \dots$

A simple cycle must pass through the root, since deleting the root leaves a path. Such a cycle uses two distinct root attachments di and dj , so its length is $d|i - j| + 2 \geq d + 2$, with equality for consecutive attachments. Finally, a non-root vertex has its at most two path neighbours and at most one root edge. $\square$

In particular W_2 is a triangle-free representation of ω whose rank is ω , the same as the birthday of its value and as the rank of $C(\omega)$. The finite pentagons raise the presentation rank above the birthday of their value, but [Proposition 11.2](#) shows that this cost is not intrinsic to large girth: at a limit stage it vanishes. This is why the finite minimum-rank question of [Remark 6.4](#) does not have an automatic transfinite analogue.

11.2 Why the root's infinite degree is not an accident

[theorem 9.5](#) bounds the maximum degree of its finite outputs by three, while the root of W_d has infinite degree. That asymmetry is forced.

Proposition 11.3 (No transfinite form has finite outdegree everywhere). *A well-founded form with only finitely many options at every descendant has finite transitive option closure. Consequently every form of transfinite rank has a vertex of infinite outdegree.*

Proof. Suppose the closure were infinite. The root has finitely many options, so at least one of them has infinite descendant closure; repeating the choice produces an infinite sequence of successive options, contradicting well-foundedness. This is the usual finite-branching argument behind König’s lemma. A form with finite transitive closure has finite rank. $\square$

So the global degree bound of [theorem 9.5](#) provably cannot extend to a representation of ω ; [Proposition 11.2](#) confines the necessary infinite degree to a single vertex, which is the best possible. No assertion is made here about arbitrary surreals of ordinal birthday beyond these examples.

12 Exact computation and reproducibility

The proofs do not depend on experimental evidence. The accompanying program is provided to make the finite examples inspectable, to check boundary cases, and to detect implementation or transcription mistakes.

12.1 Representation and arithmetic

The implementation uses Python’s standard library only. Each form is interned by the ordered pair of its sorted left and right option-ID sets. Equal-valued forms with different option sets remain distinct. All numerical values are `fractions.Fraction` objects; there is no floating-point arithmetic.

To evaluate a finite cut, the main evaluator extracts its greatest left bound and least right bound. It first checks whether zero is allowed, reflects an entirely negative interval, and tests the least admissible positive integer. When the interval contains no admissible integer, it searches successively finer dyadic grids for the first admissible fraction. In an interval confined between consecutive integers, the least denominator exponent gives the simplest value. Explicitly, for a positive interval (a, c) the least candidate numerator at scale 2^r is

$$p_r = \lfloor 2^r a \rfloor + 1, \tag{12.1}$$

which is accepted exactly when $p_r/2^r < c$; negative intervals are reflected. Strict inequalities are used at every step, and no rounding tolerance or floating-point comparison occurs anywhere.

Three separate oracles check the same objects in different ways, and all three belong in the record because they fail differently.

- (i) *A second cut evaluator.* For the exhaustive experiment, a separate evaluator enumerates values by intrinsic birthday and selects the first value satisfying every cut inequality. It does not use the main evaluator’s dyadic-grid search (12.1). The two evaluators agree on all 3,860 forms encountered in the exhaustive closure.
- (ii) *A recursive comparison oracle.* A separate implementation of (3.2) decides the order relation using only option identifiers, with no cached rationals at all. The bundled suite verifies $G \leq C(1/2)$ and $C(1/2) \leq G$, and checks all 36 ordered comparisons among their six distinct subforms against exact rational ordering. Unlike (i) it never computes a value, so it tests the order-theoretic content of this witness rather than the arithmetic kernel.
- (iii) *A graph oracle.* Girth is computed by breadth-first search from every vertex of the undirected adjacency structure built from option identifiers, and subgraph containment by the exact injective backtracking search described below. These are graph computations that use no arithmetic.

Together these are independent algorithmic cross-checks, not an independent formal verification of the entire software stack.

12.2 Why the enumeration is exhaustive

A finite dependency graph admits a topological ordering beginning with its unique sink O and ending with its root. For a fixed sequence of earlier vertices, every possible new form is described by choosing, for each earlier vertex, one of three states: absent, left option, or right option. The program examines every such choice and rejects invalid cuts. It also rejects a new vertex if that exact form has already appeared in the current sequence.

Some intermediate sequences contain vertices not used by the final root. Therefore a root is recorded at size n only when its transitive option closure has exactly n vertices. Different topological orderings of the same form are deduplicated by structural interning: the 3,860 distinct forms were obtained from 4,651 retained topological labellings. Conversely, every numerical form of size at most the bound has at least one such ordering and is therefore examined. This proves exhaustiveness for the stated finite vertex bound; it is not a heuristic search restricted to unary or binary forms.

The implementation also checks ordinary, non-induced subgraph containment by an exact injective backtracking search after forgetting directions, roots, values, and option sides.

Vertices	Distinct forms	Noninteger	Triangle-free		Canonical-subgraph failures
			Noninteger	Value $1/2$	
1	1	0	0	0	0
2	2	0	0	0	0
3	10	2	0	0	0
4	123	32	0	0	0
5	3,724	1,200	8	4	16
Total	3,860	1,234	8	4	16

Table 3: Complete enumeration through five vertices, by number of distinct subforms, with structural identity rather than equality of values. These are not counts by birthday. The last column tests the weakest unlabelled undirected subgraph relation.

12.3 Reconciling the two failure statistics

table 3 reports two different counts at five vertices, and both are real. The sixteen canonical-subgraph failures are the outcome of the broader test: they include girth-three forms of $1/4$, $3/4$, $-1/4$ and $-3/4$, which do have a triangle but not the canonical diamond, whose embedding needs two triangles sharing an edge. The eight triangle-free noninteger forms are the outcome of the narrower test, which asks only for the absence of a triangle. Filtering the sixteen certificates to those of noninteger value and girth at least four returns exactly those eight, and four of the eight have value $1/2$ (the other four have value $-1/2$). Every triangle-free noninteger form fails canonical-subgraph containment, since every noninteger canonical graph has a triangle by Lemma 10.1. The converse fails, as witnessed by the girth-three counterexamples above. Thus the eight forms satisfying the triangle-free condition form a proper subset of the sixteen containment failures.

The absence of a triangle-free noninteger form at four vertices is an independent numerical confirmation of Lemma 6.1, and the four half-valued optimal witnesses at five vertices confirm theorem 7.7 at $g = 4$. The article prints two of the four; no uniqueness is claimed.

12.4 Assertions that were actually run

For every enumerated form, the program checks the prefix path, the birthday lower bound, all three graph bounds, the value- and side-preserving spanning statement whenever the vertex count is minimal, and uniqueness whenever the edge count is minimal. It checks the enumeration formula in every one of the 31 value classes with birthday at most four. Exactly 31 enumerated forms attain the edge minimum, one canonical form for each class.

The infinite constructions are tested separately at twenty odd-cycle instances ($1 \leq m \leq 20$) and 126 spaced-spine instances. The latter consist of all 63 values of birthday at most five, each at spacings $M = 4$ and $M = 6$. Each test checks the value, rank, vertex count, edge count, cycle rank, and girth bound. The Fibonacci leaf bound of [Proposition 8.4](#) is checked at 4,096 instances spanning denominator exponents 0 through 12, together with its attainment claim. The experiment does not assert that a finite sample proves any infinite family; those proofs are in [sections 7 to 9](#).

The tree-and-spine construction of [section 9](#) has its own test suite, which is kept separate because it exercises a different code path. It checks the exact cycle family of [section 7](#) at length three and at every length five through sixty-four, giving 61 certificates, and the sparse construction at every value $x = p/32$ with $-64 \leq p \leq 64$ at requested girths 3, 4, 5, 8, 12, giving 645 grid certificates. For every case it checks the value, the degree bound, the girth, and a reconstructed certificate, and in the noninteger cases also the vertex, edge, and height formulas (9.6)–(9.8). The largest grid graph had 176 vertices and rank 166. With the four showcase examples of [table 4](#), 710 cycle or sparse-construction certificates were generated and checked. Maximum leaf counts for denominator exponents one through eight were measured as 2, 3, 5, 8, 13, 21, 34, 55, attaining the Fibonacci bound in that range. Planarity is proved in [Lemma 9.4](#); it is not separately software-tested.

Value	Requested girth	Actual girth	Leaves	Vertices	Edges	Rank	Size bound
5/8	5	7	5	30	33	27	35
−11/16	8	9	8	66	72	61	80
53/32	12	15	13	176	187	166	195
85/256	10	11	55	536	589	486	660

Table 4: Exact outputs of the tree-and-spine implementation of [section 9](#). The construction can exceed the requested girth, because the reset and padding steps of [Lemma 9.1](#) add distance. The last column is (9.6).

The run bundled with this article used Python 3.13.5 and completed with every assertion passing. The code targets Python 3.10 or newer and requires no third-party packages. Do not use Python’s `-O` flag, which disables assertions and therefore disables most of the suite. From the archive’s top-level directory, run

```
python3 code/verify.py
python3 code/verify_sparse.py
```

The first script’s default vertex bound is five. A fresh output directory can be selected with

```
python3 code/verify.py --bound 5 --output fresh_results
```

A single bundled certificate can be rechecked directly:

```
python3 -c "import json, sys; sys.path.insert(0, 'code'); \
from verify_sparse import check_certificate; \
check_certificate(json.load(open('examples/half_cycle_5.json'))); \
print('Certificate passed')"
```

and a construction can be run interactively:

```

from fractions import Fraction
from sparse_forms import sparse_form
arena, root, bounds = sparse_form(Fraction(5, 8), minimum_girth=5)
print(arena.metrics(root))
# {'value': '5/8', 'vertices': 30, 'edges': 33,
#   'height': 27, 'max_degree': 3, 'girth': 7}
arena.save(root, 'my_5_8_certificate.json')

```

In these outputs `height` is the form rank ρ of this article, and `girth: null` denotes an acyclic graph, that is, infinite girth.

12.5 Artifacts and certificates

The module `code/surreal_forms.py` contains the exact form arena, canonical construction, graph operations, the odd-cycle and spaced-spine constructors, prefix-path extraction, and exhaustive enumeration. The script `code/verify.py` performs the checks and writes the results. The directory `results/` contains a machine-readable summary, CSV tables, the two named counterexample certificates, and all sixteen small counterexamples. The module `code/sparse_forms.py` and the script `code/verify_sparse.py` are the second implementation: they contain the tree-and-spine constructor `sparse_form`, the full cycle spectrum, the recursive comparison oracle, and the reusable `check_certificate` routine, and they write to `results/sparse_*` and `examples/`.

Both implementations use the same certificate format, so certificates produced by either can be rechecked by the other's reader. A certificate has a root identifier and an array of node records in topological order. A typical record is

```

{"id": 4, "left": [0], "right": [3], "value": "1/2", "height": 4}

```

All option identifiers in a record are strictly smaller than the record's own identifier, which makes well-foundedness syntactically evident. Reconstruction from the ordered pairs of option sets alone checks that each node is well founded, is a valid numerical cut, and is structurally distinct from every previously listed node, and that the root reaches every listed node. Values and heights in the file are therefore recomputed, never trusted. A certificate also carries the vertex and edge counts, maximum degree, rank, and girth; null girth means an acyclic graph. Vertex identifiers are local names and are not part of a form's identity.

The directory `examples/` holds eight certificates in this format, each with a matching Graphviz `.dot` rendering of its option graph: the C_3 , C_5 , C_6 and C_{12} representations of $1/2$ from [section 7](#) — note that `half_cycle_6` and `half_cycle_12` come from the even family (7.9) — and the four tree-and-spine outputs of [table 4](#). Graphviz is not a dependency of the tests or of the PDF build; with it installed, `dot -Tsvg examples/half_cycle_5.dot -o half_cycle_5.svg` renders one optional standalone picture.

The two pentagon certificates are small enough to print. They are the five-row tables in [tables 5](#) and [6](#), and either one alone suffices to verify the central counterexample by hand.

12.6 Limits of the computational evidence

The rational evaluator and the certificate reconstruction share one simplicity implementation within each of the two code bases. Between them they therefore test serialization, extensionality, and internal consistency rather than providing two entirely independent arithmetic kernels. The genuinely separate checks are the ones that do not use the arithmetic at all: the recursive comparison oracle (3.2) and the breadth-first girth computation. The second cut evaluator is separate in algorithm but not in language or libraries. All the unbounded statements — the all-dyadic girth theorems, planarity, the extremal lower bound, the

ID	Left-option IDs	Right-option IDs	Value	Rank
0	$\emptyset$	$\emptyset$	0	0
1	$\{0\}$	$\emptyset$	1	1
2	$\emptyset$	$\{0\}$	-1	1
3	$\{2\}$	$\emptyset$	0	2
4	$\{3\}$	$\{1\}$	1/2	3

Table 5: A complete certificate for the first pentagon (1.1). The root is vertex 4. This is entry 0 of the sixteen five-vertex counterexamples.

ID	Left-option IDs	Right-option IDs	Value	Rank
0	$\emptyset$	$\emptyset$	0	0
1	$\{0\}$	$\emptyset$	1	1
2	$\emptyset$	$\{1\}$	0	2
3	$\{2\}$	$\emptyset$	1	3
4	$\{0\}$	$\{3\}$	1/2	4

Table 6: A complete certificate for the second pentagon (1.3). The root is vertex 4. Its undirected edge set is $\{01, 12, 23, 34, 40\}$: every vertex has degree two and the graph is connected, so it is C_5 , while the canonical half has three mutually adjacent vertices. This is entry 3 of the same sixteen-element certified list, so the two witnesses are demonstrably different members of one machine-checked enumeration.

classifications, and the two ω propositions — are established by the written proofs, not inferred from any finite test set.

13 Interpretation, limitations, and further questions

13.1 What has been settled

The literal canonical-subgraph assertion is false. Its failure is witnessed by two extremely small numerical forms, not by a transfinite pathology or a non-number game, and the exhaustive search shows those two are among exactly sixteen five-vertex witnesses. The stronger large-girth theorems show that no fixed finite cyclic graph can be forced inside every representation of a fixed finite-birthday surreal value, and that the sparse representations can be taken planar with maximum degree at most three. For the single value $1/2$ the cost of girth is known exactly. At the same time, the prefix theorem and the exact minima reveal substantial rigidity that ordinary subgraph containment fails to capture.

It is useful to separate three types of necessity. Prefix *values* are unavoidable. Certain amounts of graph *complexity*, measured by vertex count and cycle rank, are unavoidable. Individual canonical *edges* are unavoidable only under the additional vertex-minimality hypothesis proved here. The distinction explains how a fixed dyadic denominator can force a positive cycle rank while allowing all short cycles to be eliminated.

Recent work by Roughan studies the generation of random surreal forms for benchmarking and emphasizes graph structure as a source of varied inputs [4]. The constructions here provide deterministic complementary test families: a fixed value with a prescribed long odd cycle, fixed values with arbitrarily large girth, and complete sets of vertex-minimal forms. These are mathematical input families, not measured claims about the running time of an external implementation.

13.2 A proof audit

The logical dependencies are short and worth stating explicitly. The two five-node counterexamples use only the evaluation of cuts with values $0, \pm 1$ and $1/2$, structural identity, and the absence of a triangle in a pentagon; their mathematical content can be checked without any general construction and without any code.

The spaced-spine theorem 8.1 uses the canonical occurrence tree, the zero spine (7.1), the rank formula (8.8) with the spacing condition $2M > b$, and the observation that a cycle in a tree joined to a path must traverse a spine segment between attachment vertices. The tree-and-spine theorem 9.5 uses four different facts: a finite canonical cut tree can be stopped at integer leaves; unary cuts create distinct integer-valued markers arbitrarily far apart; replacing an option by an equal-valued form preserves the value of a cut; and the same cycle observation. In each construction the non-collapse lemma — the rank separation inside the proof of theorem 8.1, and Lemma 9.3 respectively — is indispensable: without it the graph argument would describe allocated objects rather than mathematical forms.

The classifications use the girth theorems only in the negative direction. theorem 10.2 obtains its positive half from prefix occurrence; theorem 10.3 obtains its stronger positive half from recursive comparison alone. The exact extremal function 7.7 uses the two odd families, the even family, the forest obstruction, and the complete four-vertex analysis 6.1. The two ω results are independent of each other and of the finite constructions. None of these arguments uses a conjectural estimate, a numerical approximation, or an extrapolation from the finite enumeration.

13.3 What has not been claimed

The original source and the later benchmarking paper were inspected, together with sign-expansion and Dali references. The target question was read in the 2018 arXiv version, at the caption of Figure 3 on printed page 7. A 2023 journal version of the same paper is indexed in *Mathematics Magazine* 96, pages 329–343 [3]; its full text was not retrievable in this research session, so the quoted wording is attributed to the accessible preprint and is not asserted to occur unchanged in the journal version, nor is it asserted that the journal version contains no answer. Searches for the canonical-subgraph question, counterexamples, and girth constructions did not locate an earlier resolution on 20 September 2026. Search coverage and publisher access were limited. This supports treating the documented question as a research target, but it does not certify the novelty of every theorem or exclude an answer in an unindexed source or private correspondence. The prefix argument is a direct application of standard simplicity and is presented with its proof, rather than advertised as a new foundational principle.

No Lean or other proof-assistant formalization is included. The exhaustive program checks a finite universe and explicit finite instances. The ordinal arguments, the classifications, and the infinite families are justified by the written mathematical proofs. None of the results is a theorem about all combinatorial games: the numerical-cut condition is used essentially.

13.4 Two questions answered along the way

Two questions that each half of this investigation had left open are settled by the other half, and we record the resolutions rather than reprinting the questions as open.

First, whether a representation whose rank equals the intrinsic birthday of its value must contain the canonical option graph: Proposition 6.3 answers this negatively at $x = 1/4$, under the unlabelled convention, by a rank-minimal form whose graph has one triangle where the canonical diamond needs two. See Remark 6.4. Second, the optimal girth cost at a fixed value: theorem 7.7 determines it exactly for $x = 1/2$, where the answer is g vertices for every integer $g \geq 5$. See Remark 7.8. The counterexample also defeats stronger subgraph conventions; minor containment and restricted presentations require separate analysis. The optimal girth cost remains open here for general dyadics.

13.5 Five questions left by this analysis

Question 13.1 (Cycle rank versus girth). For every dyadic x with $\kappa(x) = k$, can arbitrarily large girth be achieved while keeping $\beta = k$?

The answer is yes for integers and for $x = \pm 1/2$ by the explicit path and odd-cycle constructions. Neither uniform construction answers the question in general: the spaced-spine construction has cycle rank $L - 1$ and the tree-and-spine construction has cycle rank $\ell - 1$, and both may exceed k . This is a natural question arising from the present proofs, not a claim that its status has been exhaustively surveyed.

Question 13.2 (Minor and subdivision replacements). Which canonical dependency graphs are forced as undirected minors or topological minors of every equivalent finite form?

Large girth alone is not an obstruction to either kind of containment. The prefix-path theorem supplies overlapping reachability witnesses, not the branch-set or disjoint-path conditions needed for these stronger conclusions.

Question 13.3 (Optimal girth cost). For a fixed noninteger dyadic x , what is the least number of vertices needed for a representation with girth at least g ? Can the bound (8.11) be improved uniformly in the denominator exponent?

For $1/2$, [theorem 7.7](#) gives the exact answer g for $g \geq 5$. For general x , both proofs give a bound linear in g with an explicit value-dependent constant, since M may be chosen proportional to $g + b(x)$ and d equals $g - 2$. The Fibonacci estimate quantifies those constants but does not establish an optimal dependence on k .

Question 13.4 (Optimal sparse complexity, height included). For a fixed noninteger dyadic x , determine also the least form rank achievable subject to girth at least g .

This is the companion of [Question 13.3](#) in the second size parameter. [theorem 7.7](#) settles the vertex count for $x = 1/2$ and says nothing about rank; (8.6) and (9.7) give upper bounds only, and the two constructions disagree about which is smaller.

Question 13.5 (Restricted arithmetic presentations). Which canonical subgraphs are forced if the alternative forms are required to be outputs of a specified collection of raw arithmetic operations on canonical inputs, rather than arbitrary valid cuts?

Both witnesses here are valid cut constructions, and nothing in either proof identifies them as outputs of any particular restricted arithmetic syntax. Such a restriction would define a different problem, and it is the one direction in this list that changes the class of admissible representations rather than the notion of containment or the size measure.

14 Conclusion

A representation of $1/2$ need not contain its canonical triangle: the five forms in (1.1) instead form a pentagon, and so, differently, do the five forms in (1.3). This gives a direct, fully finite negative answer to the selected published question, twice over. The answer can be strengthened in two complementary directions. On the flexible side, representations of every finite-birthday value can avoid all cycles below an arbitrary length, and can be taken planar with maximum degree at most three; at $1/2$ the exact price of girth at least g is g vertices for integers $g \geq 5$. On the rigid side, canonical prefix values cannot disappear, and the least vertex count, edge count, and cycle rank are exactly calculable from the birthday and denominator.

The sharp replacement for universal edge containment is therefore not that canonical structure vanishes, but that it survives in a different form: as prefix reachability, as minimum graph complexity, and as a mandatory spanning graph precisely at minimum vertex size. The transfinite example shows why even the reachability formulation must be handled carefully at limit birthdays.

A Verification pseudocode

The exhaustive search is summarized below. `intern` identifies only structurally equal forms. The actual implementation includes strict numerical validity checks, exact values, ranks, and deduplication of recorded roots.

```
search(prefix):
    root = last(prefix)
    if option_closure(root) contains every vertex of prefix:
        record root at size len(prefix)
    if len(prefix) == vertex_bound:
        return
    for choice in {absent, left, right} ^ len(prefix):
        L, R = option_sets(prefix, choice)
        if every value in L is strictly below every value in R:
            new = intern(L, R)
            if new not in prefix:
                search(prefix followed by new)

search([empty_form])
```

The finite prefix-path procedure is constructive as well. It uses exact comparisons and the fact proved in [Lemma 4.2](#) that an eligible option must exist.

```
path = [root]
current = root
for q in proper_sign_prefixes(value(root)), longest first:
    while value(current) != q:
        if q < value(current):
            choose a left option B with q <= value(B) < value(current)
        else:
            choose a right option B with value(current) < value(B) <= q
        append B to path
        current = B
return path
```

In a finite graph, ranks strictly decrease and ensure termination. In the transfinite proof, ordinal rank induction replaces any proposed numerical iteration bound. The code implements only the finite algorithm.

B The tree-and-spine construction in pseudocode

The following summarizes the mathematical construction of [section 9](#). It is not the code used as evidence; the executable implementation is `code/sparse_forms.py`.

```
SPARSE(x, g):
    require x dyadic and g >= 3
    if x is an integer:
        return the canonical unary chain for x

    T := unfolded canonical binary cut tree for x,
        stopping each branch at its first integer
    d := g - 2
    spine := [empty form]
    markers := []

    for each integer leaf label n of T, in left-to-right order:
        reset the current integer value to 0 if necessary
```

```

repeat ceil(d/2) times:
    append a unary left cut    # value 0 -> 1
    append a unary right cut   # value 1 -> 0
    append abs(n) unary cuts toward the integer n
    record the current form as this leaf's marker

rebuild every internal node of T as {left_child | right_child}
return the rebuilt root, interning by complete option sets

```

The reset and padding steps are what make the actual girth of the output exceed the requested girth in table 4. Interning by complete option sets, rather than by value or by allocation, is what Lemma 9.3 justifies.

References

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